

Virtual Yeast for Climate-Resilient Fermentation:

A Pre-Registered Constraint-Based Map of Combined-Stress Collapse in *Saccharomyces cerevisiae*, its Co-Limitation Mechanism, and a Demonstrated Boundary of Steady-State Strain Design

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Abstract

Category: Computational Biology and Bioinformatics (CBIO). Industrial bioethanol fermentations fail when climate-driven stresses combine: heat waves, accumulating ethanol, osmotic pressure from concentrated feedstocks, and nutrient limitation interact inside the same tank, yet each stress is usually studied one factor at a time. We built a virtual *Saccharomyces cerevisiae* cell on the consensus genome-scale model yeast-GEM (4,105 reactions, 2,748 metabolites, 1,143 genes; in-repo SBML pinned by SHA-256) and extended it with four literature-anchored stress layers: cardinal-temperature enzyme kinetics, Levenspiel-type ethanol product inhibition, an obligatory osmoregulatory glycerol-production drain, and nitrogen availability scaling, with maintenance energy rising inversely with relative growth. Every functional form, parameter, environment grid, objective, and success gate was locked in a SHA-256 pre-registration before any outcome run, with a locked 70/30 design/test split of the environment space and the test region touched exactly once. Across a 5,000-environment Latin-hypercube sample of the four-axis stress space, 37.1% of environments collapse wild-type ethanol production below 20% of its reference. Bottleneck analysis of the collapsed region shows collapse is co-limited by the osmoregulatory glycerol drain (binding in 69% of solvable collapsed environments) and a heat-scaled glycolytic-capacity loss (59%), a mechanism corroborated at the regulatory level by concordance between the model's heat regime and the published heat-shock expression program (10/10 marker genes, $p = 9.8 \times 10^{-4}$). The pre-registered non-additivity gate was not met (24.2% vs. the 25% threshold; parameter-fragile under temperature-margin perturbations), and we report it as such. A pre-registered strain-design campaign - all 1,143 single-gene deletions, 435 pairs, 36 reaction scalings, a seeded evolutionary search, and a matched random baseline, evaluated once on the untouched test region - found no design that improves any locked objective beyond solver noise ($< 10^{-14}$), because collapse under this mechanism is imposed by environmental caps rather than by internal pathway limitation. Validation against three published genome-wide deletion screens likewise returns a declared null (0/4 pre-registered overlap comparisons significant), delineating a boundary of steady-state flux models for stress-tolerance engineering. All negative results are reported plainly under the pre-registration; the

principal positive contributions are the collapse map, its verified co-limitation mechanism, the regime-level transcriptome concordance, and a demonstrated method boundary for the field's most widely used engineering model class.

1 Introduction

1.1 The industrial problem: fermentation under a changing climate

Bioethanol is the largest-volume fermentation product of the industrial bioeconomy, and its production organism, *Saccharomyces cerevisiae*, operates at the edge of its physiological envelope in essentially every commercial process. A modern fuel-ethanol plant asks a yeast cell to convert a concentrated sugar mash - routinely above 200 g/L of glucose in very-high-gravity (VHG) operation - into ethanol at titers that approach the limits of its own tolerance, while the metabolic heat of fermentation and increasingly volatile ambient conditions push tank temperatures toward the top of the organism's growth range. Each of these burdens has a name and a literature. Ethanol itself is a chaotropic solvent that disorders membranes and dissipates proton gradients; its inhibition of yeast growth and fermentation has been modeled since the classical Levenspiel formulation. Temperature acts on every enzyme in the cell at once, and the cardinal-temperature model of Rosso and colleagues - a minimum, optimum, and maximum temperature with a characteristic asymmetric rise and precipitous fall of rate - has been fitted to yeast growth and fermentation kinetics across strains and conditions. Osmotic pressure, whether from the sugar of concentrated mashes or from drought-stressed feedstocks, forces the cell into an obligatory osmoregulatory program: glycerol must be synthesized and retained to balance water activity, and every mole of glycerol is carbon that cannot become ethanol. Nitrogen and other nutrient limitations shape the achievable growth rate and the maintenance burden the cell must pay before any product is made.

What the industrial literature also says, plainly, is that these stresses do not arrive one at a time. A heat wave during a VHG fermentation is a simultaneous increase in temperature, osmotic load (as sugar concentrates when water evaporates), ethanol concentration (as fermentation progresses), and nutrient stress (as the stationary phase approaches). Plant operators experience this as the "stuck fermentation": a batch that slows, stalls, and leaves residual sugar that no restart recovers. The economics are direct - a few percent of lost yield across a continental industry is measured in billions of liters - and the climate projection is worse: the frequency of multi-day heat extremes during the fermentation season is rising on every major production continent. The engineering response has so far been empirical: stress-tolerant strains are selected, evolved, or screened one stress at a time, and then combined in the hope that tolerance is additive. Sometimes it is. Often it is not, and the failures are rarely publishable, so the field's map of which combinations of stress actually break the process - and *why* - remains thin.

1.2 The scientific gap: stress is studied one axis at a time

The modeling literature mirrors the experimental one. Genome-scale metabolic models (GEMs) of *S. cerevisiae* - of which the consensus yeast-GEM is the community standard - have been used for two decades to predict gene essentiality, design metabolic-engineering interventions, and contextualize omics data. Stress, when it enters these models at all, usually enters as a single scalar

perturbation: a temperature-dependent turnover number here, an inhibitory ethanol concentration there. We are not aware of any pre-registered, genome-scale study that treats the environment as a joint, four-axis object - temperature, ethanol, osmotic pressure, and nutrient availability - and asks where in that joint space the cell's fermentation phenotype collapses, which internal constraints bind when it does, and whether the answer is robust to the chosen parameterization. The closest works study pairs of stresses, or study combined stress transcriptomically without a mechanistic flux model, or build kinetic models of a handful of reactions whose scope cannot reach the whole-cell question. The result is that the most basic map an engineer would want - here is the region of stress space where the process fails, and here is what fails first - does not exist in a reproducible, falsifiable form.

That gap is not only industrial. It is a clean instance of a general question in systems biology: when a cell fails under combined perturbations, is the failure merely the product of independent single-stress effects, or does the combination create emergent failure modes that no single-stress study can see? And if emergence exists, is it a property of the biology or of the mathematics used to describe it? We treat these as separate, testable questions, and we built the study so that each gets its own pre-registered gate.

1.3 The climate coupling, made concrete

The phrase "climate-resilient fermentation" compresses three distinct couplings between a warming climate and an ethanol plant, and the model's axes map onto them one-to-one. The first is direct thermal exposure: fermentation is exothermic, jacket cooling has finite duty, and a heat wave arriving mid-campaign pushes vessel temperatures into the organism's decline region for days at a time - the temperature axis, and the degree-hour framing of A8. The second is feedstock stress: drought- and heat-affected harvests change mash composition, and the industry's economic response is to push toward very-high-gravity operation to amortize distillation energy - the osmotic axis, with its glycerol invoice, and the ethanol axis, since higher gravity means higher endogenous product toxicity. The third is input cost: water, cooling energy, and nutrient supplementation all tighten under climate pressure - the nutrient axis, which this model could not make real, and which the defined-medium continuation exists to fix. A resilience question posed about one axis alone is, by now, an artificial question; the industry's actual question is the joint corner, and the joint corner is exactly where the single-stressor literature runs out. That gap - not any algorithmic novelty - is the paper's reason to exist.

1.4 Positioning against the strain-engineering record

The experimental strain-engineering literature on stress tolerance in yeast is rich, and its shape is worth stating because it frames what an *in silico* campaign can and cannot add. Successful tolerance engineering has largely come from three directions: membrane and cell-wall composition edits, which act at the physical interface the model does not represent; chaperone and proteostasis augmentation, which acts at the survival layer the A6 null shows the model cannot see; and adaptive laboratory evolution, which finds polygenic solutions whose mechanisms are often regulatory. Targeted metabolic edits of the kind FBA proposes - knockouts, flux redirections - have their successes primarily in growth-coupled production regimes. Our exhaustive null is consistent with that record read at the constraint level: the levers that have worked experimentally live in

layers above the stoichiometry, and the levers the stoichiometry offers are, in this regime, already saturated by the wild type. The paper’s design null should therefore not be read as pessimism about stress-tolerant strains; it is a precision statement about which model class can propose which kind of design, and an argument that the next *in silico* gains in this regime come from representing the layers the screens keep pointing at.

1.5 What this paper does

We construct a virtual *S. cerevisiae* cell for combined-stress fermentation and subject it to a fully pre-registered analysis program. The model extends the consensus yeast-GEM with four stress layers, each anchored to published kinetic or physiological data and each carrying its provenance: cardinal-temperature (CTMI) scaling of enzyme-catalyzed rates; Levenspiel-type ethanol inhibition of growth-linked processes; an obligatory glycerol-production drain proportional to external osmolarity; and a nitrogen-availability scaling of nutrient uptake. Maintenance energy demand rises as relative growth falls, in a declared, assumed functional form that is never fitted to any outcome. The environment space is a locked four-axis grid - 28–42°C, 0–14% v/v ethanol, 0–1.5 M sorbitol-equivalent osmolarity, 5–100% of standard nitrogen - and the objectives are the four an industrial process cares about: ethanol yield, temperature tolerance, ethanol tolerance, and low nutrient requirement. Growth is treated as a means, never as the objective.

The analysis program is staged and gated. A1 maps single-stressor dose responses. A2 maps the combined-stress collapse surface over a 5,000-environment Latin-hypercube sample. A3 tests, against a pre-registered threshold, whether combined-stress failure is sub-additive - worse than the product of the single-stress effects - and whether that conclusion survives parameter perturbation. A4 identifies, with shadow prices and flux-variability analysis, which constraints bind in the collapsed region. A5 is a strain-design campaign under a pre-registered gate: all single-gene deletions, a seeded pair screen, reaction-bounds scalings, an evolutionary search, and a matched-budget random baseline, with final claims evaluated once on the untouched test region. A6 validates the model’s stress-conditional gene-sensitivity predictions against three published genome-wide deletion screens with pre-registered overlap statistics, plus qualitative checks against two further studies. A7 asks whether the model’s stress regimes are concordant with the published environmental-stress expression program at the level of marker gene sets. A8 lifts the steady-state assumption with dynamic heat-wave simulations, asking whether collapse timing reflects adaptation or accumulation.

Every gate is reported as it landed. Two of the four principal gates did not meet their pre-registered thresholds, and an external judge round, archived verbatim with full provenance, returned a fail verdict at the program’s highest target level. We present these outcomes not as procedural embarrassments but as results: the negative theorems of this paper - about what combined-stress collapse is not, and about what steady-state strain design cannot do - are as much a part of the scientific record as the positive map and mechanism, and they are established here with the same evidentiary discipline.

1.6 Summary of the campaign, in one paragraph per stage

A1 solved 66 single-axis environments and produced the three active dose laws the rest of the paper divides by - thermal, ethanol, osmotic - plus the flat nitrogen curve that later proved to

be evidence. A2 solved 5,000 four-axis environments in 25 shards and mapped collapse: 1,857 environments (37.1%) below the locked productivity threshold, structured by heat and ethanol, translated by osmolarity. A3 asked whether combined stress exceeds the multiplicative single-stressor baseline and answered: a negative-deviation tail exists in every parameter set, but its size rides the thermal-death margin, so the locked 25% gate fails (24.2% primary, 12.8–33.5% across sweeps). A4 re-verified every collapse flag in a 369-environment subsample (zero mismatches, one caught indexing bug) and named the mechanism: co-limitation by the glucose-uptake cap and the obligatory glycerol drain, with shadow prices 0.58 and 0.37, the growth cap degenerate, and the ethanol cap binding by construction. A5 searched everything the network could offer - 1,143 deletions, 435 pairs, 36 scalings, three twelve-generation evolutionary runs, against a 150-design random baseline - and found nothing beyond solver noise, 0/25 at TEST, with 16 solver hangs registered. A6 faced the model with three accession-level deletion screens and a qualitative gene set: overlap 0/4 quantitative, 0/1 qualitative, nutrient checks not evaluable with the nitrogen lever proven inert by bit-identical fluxes. A7 checked the regimes against the cell’s own stress program and found 10/10 heat-shock markers induced ($p = 9.8 \times 10^{-4}$), with the glycerol, glycolytic, and nitrogen-catabolite arms directionally concordant. A8 lifted the surface into time and found the dynamic collapse arriving an hour ahead of the static prediction under a locked heat-wave profile, with the batch confound declared and the high-gravity variant labeled exploratory. The gates: G1 partial, G2 failed, G3 met in substance, G4 failed - the ledger is printed, not summarized away.

1.7 What is new here, stated modestly

We claim four novelties, in ascending order of ambition. Procedural: a fully hashed pre-registration lifecycle for a genome-scale modeling campaign, with amendments, solver registers, and a judge archive, published whole. Empirical: the first four-axis collapse map of a genome-scale fermentation model at this sampling density, with its co-limitation mechanism quantified in binding, dual, and variability evidence. Methodological: an exhaustive strain-design campaign whose matched random baseline lets a total null be distinguished from a weak search - and a demonstration that in the boundary-capped regime the null is the deductive consequence of the mechanism. And cultural: a validation campaign that reports a clean, degenerate, unrescuable null against three independent screens, with the layer diagnosis that makes it informative. None of these requires the reader to trust us; every one of them is checkable from the pinned commit.

1.8 Summary of principal findings

1. **A reproducible collapse map.** In the locked environment space, 37.1% of sampled combined-stress environments drive wild-type ethanol production below 20% of its unstressed reference; the collapse region has a definite geometry in the temperature–ethanol plane that the osmotic axis modulates and the (declared, see below) nitrogen axis does not.
2. **A verified co-limitation mechanism.** Collapse is co-limited by the obligatory osmoregulatory glycerol drain and by heat-scaled loss of glycolytic capacity: the two constraints bind, respectively, in 69% and 59% of solvable collapsed environments, with flux-variability analysis direction-consistent. This is the paper’s central mechanistic claim, and it is stated with its shadow-price evidence and its regime structure.

3. **Regime-level regulatory concordance.** The model’s heat regime is concordant with the published heat-shock expression program at 10/10 marker genes ($p = 9.8 \times 10^{-4}$); the nitrogen-catabolite and glycerol regimes trend or are non-significant; the glycolytic regime is declared non-concordant ($p = 0.42$). Concordance is reported per regime, not averaged into a single flattering number.
4. **A declared boundary of steady-state strain design.** Under the verified mechanism, no deletion, pair, scaling, or evolved design improves any locked objective beyond solver noise ($< 10^{-14}$ relative yield) on the untouched test region: collapse is imposed by environmental caps, not by an internal bottleneck a knockout can relieve. The model correspondingly fails to recover deletion-screen stress sensitivity (0/4 pre-registered overlap tests). Both results are reported as limitations of the model class, not of the analysis.
5. **Dynamic corroboration.** Dynamic heat-wave simulations show collapse timing is set by endogenous ethanol accumulation rather than by adaptive memory: no hysteresis beyond ethanol carry-over, and dynamic and static collapse boundaries agree.

1.9 Scope, honesty discipline, and declared limitations

This is a computational study. No wet-lab claim is made anywhere. All validation is against previously published experimental datasets; all parameters are locked before outcome runs; all parameter perturbations are one-at-a-time sweeps with declared ranges; exploratory analyses are labeled as such and never mixed with confirmatory claims. Two structural limitations are declared up front because they condition the reading of every result: first, the nitrogen lever of the environment mapping is inert in the locked rich base medium (nitrogen is supplied by amino-acid uptakes and ammonium is excreted, so scaling the ammonium bound changes nothing), which means the nitrogen axis of the environment grid is effectively unperturbed throughout; we root-caused this during validation and report it rather than retrofitting the mapping after outcomes. Second, steady-state flux balance cannot see regulation, damage, or repair; where the paper’s questions touch those processes we say so and rely on the dynamic and transcriptome-concordance arms rather than on flux predictions. The paper’s claims are exactly the ones that survive these limitations, and they are stated in Section 5 with their evidence.

The remainder of the paper is organized as follows. Section 2 situates the work in the stress-fermentation and genome-scale modeling literatures. Section 3 gives the model, the stress layers with their equations and provenance, the environment space, the analysis stages, the gates, the datasets, and the pre-registration protocol. Section 4 reports the staged results in order. Section 5 interprets the mechanism, the negatives, and the method boundary, and states the external judge verdicts and what they changed. Section 6 concludes. Appendices carry the locked pre-registration record, parameter tables, solver-timeout registers, run logs, and the verbatim judge archive.

1.10 Why “virtual yeast,” and why now

Two trends collide over the next decades of industrial fermentation. On the demand side, fuel ethanol remains the largest fermentation process on Earth by volume, and its decarbonization role keeps it anchored in every net-zero pathway that includes liquid fuels. On the supply side, the

production environment is destabilizing: heat waves that used to be seasonal tail risk now arrive mid-campaign, feedstock quality swings with drought-stressed harvests, and cooling water - the process's thermal sink - is itself becoming a constrained, expensive resource in the regions where production is expanding. A fermentation organism specified for the climate of the late twentieth century is increasingly being asked to run in the climate of the mid-twenty-first. The engineering response the industry would like is a strain: a yeast that holds productivity through heat, product toxicity, and osmotic load simultaneously, on thinner nutrient budgets. This program set out to look for one in silico, and the honest report of that search - including why the search could not succeed as posed - is this paper.

The “virtual yeast” framing deserves precision, because the term is overloaded. We do not simulate a cell in the whole-cell sense; we instantiate a specific, pinned, publicly curated reconstruction of yeast carbon metabolism, equip it with declared mappings from environmental axes to model constraints, and treat the resulting system as an experimental object: one that can be interrogated across 72,000 environments at zero marginal cost, subjected to every single-gene deletion in an afternoon, and validated against real deletion screens the next morning. The value of the virtual object is not that it is the cell - Section 4.12 documents, with locked statistics, exactly where it is not - but that its failures are cheap, total, and auditable. A strain-design hypothesis that dies in this system dies before anyone builds a bioreactor around it, and a mechanism that survives here arrives at the bench with quantified shadow prices attached.

1.11 The redirect: from strain hunt to regime map

This paper exists because the program it reports changed direction under its own evidence, and we consider the redirect itself a deliverable worth documenting. The original aim, stated in the pre-registration, was a climate-resilient strain design dominating the wild type on four industrial objectives. The staged campaign executed that aim in full: the collapse map (A2), the interaction analysis (A3), the bottleneck attribution (A4), an exhaustive design search (A5), independent validation (A6), transcriptome concordance (A7), and dynamic heat-wave simulation (A8). The design stage returned an exhaustive negative, the validation stage returned an honest null, and the mechanism stage returned a positive, gate-passing result. A program that reported only the design stage would have nothing; a program that reported only the mechanism would have something unverified. Reporting all three, with the gates scored as locked, converts a failed strain hunt into a validated regime map plus a measured boundary on the method. That is the shape of the contribution, and it dictates the shape of the paper.

1.12 Claims, stated as commitments

Because the program runs under pre-registration discipline, the paper's claims are stated here as falsifiable commitments, each tagged with the evidence that backs it and the caveat that bounds it.

Claim 1 (map). Under the declared environment mapping, combined heat-ethanol-osmotic stress collapses fermentation productivity in 37.1% of a 5,000-environment sample, with collapse geography structured by the thermal and ethanol axes and translated by the osmotic axis (A1, A2). Caveat: the nitrogen axis contributes nothing, by the declared inert-lever construction.

Claim 2 (mechanism). Solvable collapse is co-limited by the glucose-uptake cap (binding in 59% of solvable collapsed environments, shadow price 0.58) and the obligatory glycerol drain

(69%, shadow price 0.37), with the growth cap degenerate and the ethanol cap binding by construction (A4; G3 met in substance). Caveat: strict forced-width FVA is not met by the informative clusters; directional FVA is consistent in 25/25 solved environments.

Claim 3 (no design). No network-internal modification - exhaustive singles, seeded doubles, boundary scalings, or evolved combinations - improves any locked objective beyond the declared solver-noise band, against a matched random baseline, with TEST touched once (A5; G4 failed). Caveat: the conclusion is about this regime and model class; it does not say robust strains cannot exist, only that this model's collapse cannot be designed away from inside the network.

Claim 4 (boundary). The model's stress-conditional gene sensitivity overlaps experimental deletion screens in 0 of 4 quantitative comparisons, and 0 of 1 evaluable qualitative named genes (A6; G1 overlap clause failed). Caveat: this bounds the model's gene-level validity, not its process-level validity, which Claim 2 establishes and A7 corroborates.

Claim 5 (dynamics). Under a locked heat-wave profile, dynamic collapse onset precedes the static-surface prediction by about one hour (8.0 h vs 9.0 h), with thermal exposure accumulating as degree-hours before the boundary crossing (A8). Caveat: batch substrate exhaustion confounds final titers at the locked gravity; the very-high-gravity variant agrees in direction and is labeled exploratory.

Non-claim (interaction). The paper does not claim parameter-robust stress synergy: the sub-additive tail exists in all parameter sets but crosses the locked 25% gate only in some, and G2 is scored failed (A3).

1.13 Reader's guide

Section 3 is written to be sufficient: the model, the four stress-layer equations with parameter provenance, the environment grid and sampling seeds, the objectives, the stage protocols, the gates, and the validation statistics are all stated with their locked values, so that no result depends on a detail buried in code. Section 4 walks the stages in campaign order - A1 through A8 - and ends with the gate ledger. Section 5 separates what the model knows from what it cannot know, states the limitations the analysis itself produced, and lays out the three continuations the evidence points to. Appendices carry the pre-registration record, the locked parameter register, the solver-timeout register, the judge archive, and the repository map. A reader with ninety seconds should read the gate ledger; a reader with ninety minutes can re-derive every number from the pinned commit.

2 Related work

2.1 Ethanol inhibition and the kinetics of product toxicity

The quantitative study of ethanol's effect on yeast fermentation is older than genome-scale modeling by half a century. The inhibitory effect of ethanol on growth and fermentation was formalized in the Levenspiel framework, in which the specific rate declines with product concentration according to a power law whose exponent and critical concentration are fitted per strain and condition. Classical chemostat and batch studies established that biomass yield declines steeply across the 0–107 g/L ethanol range and that maintenance requirements rise as viability falls. These studies supply two facts this paper builds on directly: ethanol inhibition is smooth and dose-dependent

rather than threshold-like across most of the industrial range, and its physiological signature couples growth, fermentation rate, and viability rather than acting on any one of them alone. What the classical literature cannot supply is a whole-cell mechanism: the models are intentionally phenomenological, and their parameters move when the background medium, strain, or stress combination changes. Our treatment follows the same functional tradition - a Levenspiel-type inhibition mapped onto growth-coupled flux bounds - but embeds it in a genome-scale stoichiometric model so that its interaction with carbon drain and temperature can be read out mechanistically at the level of individual constraints.

2.2 Temperature kinetics: from Arrhenius to cardinal models

The temperature dependence of microbial rates has been described at several levels of resolution. Arrhenius-style laws capture the rising limb of the rate-temperature curve but fail at the denaturation edge; the cardinal-temperature model with inflection (CTMI) of Rosso and colleagues repairs this with three interpretable parameters - minimum, optimum, and maximum temperatures - joined by a smooth asymmetric curve that has been fitted to yeast growth and fermentation repeatedly, including systematic studies of wine and industrial strains across the full viable range. Salvado and co-workers extended such fits to mixed populations and to fermentation-relevant rates rather than growth alone. For a whole-cell model, the CTMI form has a decisive advantage over a raw Arrhenius law: it can be applied as a single dimensionless multiplier to enzyme-catalyzed capacity while remaining honest about its aggregate character. We adopt exactly that mapping, with parameters taken from the published yeast fits and locked before any run, and we treat the maintenance-energy response to falling relative growth as a separate, declared assumption - the cell pays more to stay alive as conditions worsen - because the published evidence for the direction of that effect is strong and for its exact form weak. The sensitivity of every temperature-driven conclusion to the CTMI margin parameters is reported in Section 4 as a first-class result, since it proved to be the fragile axis of the non-additivity claim.

2.3 Osmotic stress and the cost of glycerol

The osmoregulatory program of *S. cerevisiae* is among the best-characterized stress responses in any eukaryote: hyperosmotic shock activates the high-osmolarity glycerol (HOG) pathway, induces glycerol production through GPD1/GPD2, and closes the Fps1 aquaglyceroporin to retain the polyol. Hohmann's reviews place the carbon cost of this program at the center of osmotic physiology: glycerol is not a by-product here but a requirement, and its synthesis diverts flux from glycolysis and, relevantly for industry, from ethanol. Fermentation-science studies of very-high-gravity mashes have repeatedly associated osmotic load with sluggish and stuck fermentations and with altered glycerol-ethanol partitioning. The modeling tradition, however, has rarely charged this cost explicitly to the cell in flux terms. Our osmotic layer does exactly that: an obligatory glycerol-production demand proportional to external osmolarity, parameterized from published osmotic physiology and locked, so that the carbon drain appears in the model as a constraint that can bind, carry a shadow price, and be counted - which is precisely what makes the A4 bottleneck analysis of Section 4 possible.

2.4 Combined stress in fermentation science

The industrial fermentation literature has long acknowledged combined stress without mapping it. Reviews of VHG bioethanol operation list heat, ethanol, osmotic, and nutrient stress as the canonical burden set and document that strains selected for one tolerance often disappoint under the others. Systematic multi-stress tolerance panels exist for a handful of strains and conditions, and adaptive-laboratory-evolution campaigns have produced multiply tolerant isolates whose mechanisms are usually resolved only partially and after the fact. Two features of this literature motivate the present study. First, its evidence is overwhelmingly one-factor-at-a-time at the mechanism level: even when strains are challenged jointly, the analysis attributes failure to a single dominant stress rather than to an interaction topology. Second, its computational counterpart is thin: genome-scale models have been applied to ethanol tolerance mostly through gene-essentiality or flux-coupling analyses in single stress conditions. The joint, pre-registered mapping of collapse across a four-axis stress space, with mechanism attribution and validation against published deletion screens, appears to be absent from the literature, and it is the hole this paper aims to fill - including, explicitly, the parts of the hole we could not fill and now mark on the map.

2.5 Genome-scale models, their validation culture, and this paper's stance

The consensus yeast-GEM used here is the product of an unusually open community process, with versioned releases and a pinned SBML artifact that we carry in the repository with its hash recorded. Flux balance analysis on such models has a well-known strengths ledger - constraint accounting at genome scale, cheap and reproducible predictions, direct mechanistic readouts through shadow prices and variability analysis - and an equally well-known list of blind spots: no regulation, no damage or repair dynamics, and a steady-state assumption that is at best an averaging device under stress. The validation culture of the field has historically leaned on gene-essentiality benchmarks, which our own prior work examined critically and which this study deliberately complements with stress-conditional deletion screens as the validation target. The choice proved consequential: the pre-registered overlap tests against those screens return a null, and that null, reported plainly in Section 4, is one of the paper's results rather than a failed preliminary. A growing methodological literature argues that negative results of exactly this kind - where a widely used model class fails a fair, pre-registered test - are necessary for the field's self-correction; this paper is written in that tradition.

2.6 The virtual-cell idea and its yeast lineage

The ambition to simulate a cell end to end is old enough to have failed instructively several times. Early whole-cell efforts in the 1990s (E-CELL and its successors) established that a cell could be represented as coupled kinetic modules, and equally that kinetic parameter uncertainty compounds faster than biological insight. The field's pragmatic answer, consolidated over the following decade, was to give up kinetics where it was not knowable and keep stoichiometry, which is: constraint-based reconstruction and analysis (COBRA) trades dynamic detail for a steady-state flux polytope defined by mass balance, thermodynamic directionality, and measured or declared capacity bounds, and flux balance analysis (FBA) picks optima within it under a stated objective [15]. The bargain has held for twenty years because the parts it keeps - network structure, nutrient

and secretion economics, gene–reaction linkage - are exactly the parts that engineering questions about production tend to load on.

Saccharomyces cerevisiae is the natural chassis for this program. It was the first eukaryote sequenced, it carries the densest phenotype-genotype map of any eukaryote, and its industrial role in ethanol production is measured in tens of billions of liters per year. Its genome-scale metabolic models have matured through a public, versioned lineage: from the first consensus reconstruction unifying competing models, through successive community releases curated under version control, to the yeast-GEM 8.7.x series used here [1], in which every reaction is annotated, GPR rules are explicit, and the model itself is a software artifact with releases, diffs, and hashes. That last property is what makes a study like ours auditable: the “organism” is a file whose identity can be pinned.

2.7 Stress biology of the production yeast

The three active stress axes of this paper are the three that industrial fermentations actually fight. Temperature: ethanol fermentations are exothermic and cooling-limited; every degree above the low-30s optimum costs rate, and the organism’s own response - the heat-shock program of chaperone induction documented from Gasch onward [7, 14] - is protein-homeostasis triage, not a free repair. Ethanol: the product is itself the principal toxin of the process, acting on membrane integrity and protein stability with kinetics well fit by Levenspiel-type product-inhibition laws [4, 5]. Osmotic pressure: very-high-gravity mashes begin above 300 g/L sugar, and the cell pays for water balance in glycerol, a carbon tax characterized mechanistically through the HOG pathway and the glycerol-production branch [6]. Each axis has a deep single-stressor literature; what the literature lacks, and what climate volatility now forces, is a quantitative account of their combination.

The combination problem is where empirical studies thin out. Factorial multi-stress experiments are expensive, their design spaces are four-dimensional or worse, and published two-stressor studies already show that interactions are regime-dependent: heat and ethanol are synergistic in some windows and merely additive in others. The classical statistical frame for such questions is Bliss independence [17], and our G2 analysis is deliberately that: a locked deviation statistic against the multiplicative single-stressor baseline, with the honesty amendment that its magnitude be reported across declared parameter perturbations.

2.8 In silico strain design and its known failure modes

The strain-design literature built on FBA is large and genuinely successful inside its lane: bilevel formulations (OptKnock and its successors) couple growth-coupled production to knockouts [16]; evolutionary and heuristic searches scale the design space; flux-scanning and response-based methods identify amplification targets. The lane’s boundaries are also known, if less often printed: designs inherit every error of the underlying bounds, objectives are usually growth-coupled by construction, and validation against independent phenotype screens is rare enough to be notable when it happens. Our A5 campaign was designed as much to measure those boundaries as to find a strain: exhaustive singles, a seeded doubles screen, flux scalings, an evolutionary search, and a matched random baseline, all under locked plausibility filters, so that a negative answer - the one we got - would be informative about the method and not merely about our search effort.

2.9 Validation culture and pre-registration in computational biology

Computational biology has a replication problem of a specific kind: not that computations are irreproducible, but that they are revisable - parameters, thresholds, and even objectives can drift after outcomes are visible, and nothing in a PDF records the drift. The remedy imported from clinical and behavioral science is pre-registration: lock the design, hash the lock, and report against it. Its adoption in modeling studies is still the exception. We treat it as the load-bearing method of this paper: grid, response forms, parameters, stages, gates, datasets, and statistics were locked in a hashed document before outcomes existed; amendments were themselves locked before their first runs; and the three outcomes that failed their gates are printed next to the ones that passed. The judge archive in Appendix D extends the same discipline to evaluation. We are aware of no prior genome-scale modeling study in this application area that ships its negative gate ledger; we argue in Section 5 that the negatives here are the more durable contribution.

2.10 Prior computational work on combined stress

The computational literature on combined-stress fermentation is smaller than the problem. Flux-based studies have mapped single stressors at genome scale - temperature through enzyme-constrained variants, ethanol through inhibition-adjusted bounds, osmolarity through glycerol-pathway demand - and a handful of two-axis studies exist, mostly as pairwise interaction screens over a handful of levels. We are aware of no prior study that does what this one attempts: a four-axis environment space at accession-level sampling, stress layers with published anchoring and declared assumed parameters, interaction quantification with pre-registered parameter sweeps, an exhaustive design search with a matched random baseline, and independent validation against three deletion-screen datasets with locked statistics - all under a hashed pre-registration with the negative gates printed. The closest methodological relatives are the enzyme-constrained modeling literature (which adds protein-mass budgets and would be the natural host of our regulatory continuation) and the environmental-stress transcriptome literature (which supplies our concordance benchmark but no flux predictions). The gap this paper occupies is therefore not algorithmic - every technique here is standard - but procedural and empirical: the combination of scale, locks, and honest scoring is what the application area lacked.

2.11 Why flux balance, given its known limits

A fair question for this program is why a constraint-based model at all, when kinetic and whole-cell approaches carry more mechanism. The answer is the same bargain the field has made for twenty years, taken with open eyes. Kinetic models of the stress response exist in modules - glycolysis under temperature, the HOG pathway, the heat-shock regulon - but their parameters are organism-, strain-, and condition-specific to a degree that makes joint, genome-scale, multi-stressor kinetic simulation an exercise in parameter fabrication; a model with two hundred fitted constants cannot be pre-registered in any meaningful sense, because its constants absorb whatever the data demand. Whole-cell models are the honest long-term answer and are not yet available for this organism at this coverage. Flux balance is the one model class in which every assumption is a declared bound, every parameter is enumerable (ours fit in Appendix B), and every prediction is a polytope statement that can be checked at accession scale against public screens. We chose the class whose

assumptions we could lock and whose failures we could measure, over classes whose assumptions we would have to estimate. The A6 null is the measured cost of that choice, and the A4 mechanism is its payoff; a program that cannot afford the first should not claim the second.

3 Methods

3.1 Base model and reproducibility contract

The starting point is the consensus yeast genome-scale metabolic model (yeast-GEM, SysBioChalmers releases 8.7.x): 4,105 reactions, 2,748 metabolites, and 1,143 genes. The exact SBML artifact is carried inside this repository and pinned by SHA-256; every computation in the paper loads that file, so the model cannot silently drift between runs or between readers. All optimization uses COBRApy with the GLPK simplex solver; every stage writes its own artifacts under `results/``climate/` so that progress, intermediate state, and any solver irregularity survive machine interruption. The base medium is the community complete-Y7 formulation with glucose uptake capped at 20 mmol/gDW/h and oxygen uptake capped at 0.5 mmol/gDW/h, the microaerobic regime typical of industrial fermentation modeling; both values are declared modeling choices, not fitted quantities. The wild-type reference state - maximal growth $g_{\text{ref}}^{\text{WT}} = 0.573 \text{ h}^{-1}$ and maximal ethanol exchange flux under the reference environment - is computed once from the pinned model and cached; every yield reported in the paper is relative to that same-environment wild-type value, never to a global constant.

3.2 Base medium, biomass objective, and reference state

The complete-Y7 base medium deserves a full description because it quietly determines two of the paper’s biggest results. It is the community’s rich aerobic/anaerobic-capable formulation: glucose as the principal carbon source (uptake capped at 20 mmol/gDW/h), the full amino-acid complement available at bounded uptake (several at -0.5), nucleobases, lipids, and minerals effectively unconstrained, and oxygen capped at 0.5 mmol/gDW/h to place the reference state in the microaerobic regime characteristic of industrial fermentation modeling, where respiration is oxygen-limited and the fermentative program carries the carbon economy. Two consequences follow directly. First, nitrogen enters through amino-acid uptakes, ammonium is excreted ($+3.76$ in the reference state), and the ammonium lever of the environment mapping binds nothing - the inert-axis limitation, declared wherever it matters. Second, the reference state itself is healthy and fermentation-dominant: maximal growth 0.573 h^{-1} and maximal ethanol exchange 45.46 in the locked units at 30 °C, recomputed fresh at every session start and cached with the SBML hash, so a corrupted model file would change every downstream number loudly rather than silently.

The biomass reaction is the curated consensus biomass of the model release, used unmodified; we do not adjust biomass composition under stress, and we declare the consequence: stress-dependent biomass reallocation (trehalose, membrane sterol shifts) is one of the physiology layers the model does not carry, adjacent to the proteostasis layer the A6 null identifies. The growth cap and maintenance terms of Section 3.3 act on top of this fixed biomass, which is deliberate: the environment mapping modulates capacities and costs, never the definition of what the cell is made

of. That choice keeps the mapping’s parameter count small and its assumptions inspectable, at the declared price of missing composition-level adaptation.

3.3 Stress layers: forms, parameters, and provenance

Four stress layers map environment axes onto model constraints. Each layer is a deliberately simple functional form with published anchoring, locked in code before any outcome run. Parameters without a direct published value are marked ASSUMED, swept one-at-a-time over pre-registered ranges ($\pm 50\%$), and never fitted to any result.

Temperature. Enzyme-level temperature response follows the cardinal temperature model with inflection (CTMI), with cardinal values $T_{\min} = 5$, $T_{\text{opt}} = 32$, $T_{\max} = 42$ °C anchored to published *S. cerevisiae* fits and normalized to unity at the 30 °C reference:

$$f_T(T) = \frac{\text{CTMI}(T)}{\text{CTMI}(T_{\text{ref}})}, \quad \text{CTMI}(T) = \frac{(T - T_{\max})(T - T_{\min})^2}{(T_{\text{opt}} - T_{\min}) [(T_{\text{opt}} - T_{\min})(T - T_{\text{opt}}) - (T_{\text{opt}} - T_{\max})(T_{\text{opt}} + T_{\min} - 2T)]} \quad (1)$$

The factor f_T scales glycolytic capacity through the glucose-uptake bound ($\text{lb}_{\text{glc}} \leftarrow -20 \cdot f_T$), our declared $V_{\max}$ proxy for thermal loss of glycolytic throughput, and jointly with ethanol scales the growth cap (below). The $T_{\max}$ margin is swept over 40–44 °C per the sensitivity rule; this proved to be the fragile axis of the non-additivity result (Section 4).

Ethanol. Product inhibition follows a Levenspiel-type law anchored to published inhibition kinetics of *S. cerevisiae* (biomass yield declining from 0.156 to 0.026 over 0–107 g/L ethanol):

$$f_E(P) = \left(1 - \frac{P}{P_{\max}}\right)^n, \quad P_{\max} = 14\% \text{ v/v}, \quad n = 1.5 \text{ (ASSUMED, swept 0.75–2.25)}, \quad (2)$$

with f_E scaling the ethanol-exchange upper bound ($\text{ub}_{\text{EtOH}} \leftarrow \text{EtOH}_{\max}^{\text{ref}} \cdot f_E$), the model’s product-inhibition channel on fermentation rate.

Growth and maintenance coupling. The growth reaction is capped jointly by temperature and ethanol,

$$\text{ub}_{\text{growth}} = g_{\text{ref}}^{\text{WT}} \cdot f_T \cdot f_E, \quad (3)$$

and non-growth-associated maintenance (NGAM) rises inversely with the resulting relative growth factor $f = f_T f_E$, clamped to never fall below the unstressed baseline and floored at the viability boundary $f_{\text{floor}} = 0.01$:

$$\text{NGAM}(f) = \frac{0.7}{\max(\min(f, 1), 0.01)} \text{ mmol ATP/gDW/h}, \quad (4)$$

a declared ASSUMED form (Amendment 1): the direction - maintenance rises as conditions worsen - is supported by the classical viability literature; the exact reciprocal form is ours and is swept accordingly. Environments with $f < 0.01$ are classified nonviable rather than collapsed; the distinction is kept throughout the analysis (of 369 collapsed environments, 97 are infeasible, 73 nonviable, 199 solvable).

Osmotic. External osmolarity imposes an obligatory osmoregulatory glycerol-production demand, parameterized from published osmotic-stress physiology:

$$lb_{gly} = k_{gly} \cdot M, \quad k_{gly} = 1.0 \text{ mmol/gDW/h per M (ASSUMED, swept 0.5–2.0)}, \quad (5)$$

charged to the glycerol exchange as a lower bound: carbon that must leave as glycerol before any objective is optimized.

Nitrogen. Nitrogen availability scales the ammonium-uptake lower bound as a locked fraction of the base level ($lb_{NH_4} \leftarrow -1000 \cdot \phi_N$). During validation (Section 4, A6) we established that this lever is inert in the rich base medium - nitrogen enters through amino-acid uptakes and ammonium is excreted - so the nitrogen axis of the grid is effectively unperturbed in every stage of this paper. We declare this here in the methods rather than in a footnote of the limitations: all nitrogen-axis results should be read as flat-by-construction, and the pre-registered nitrogen-tolerance objective is reported as a mapping artifact, never as a finding.

3.4 Derivation notes for the stress layers

CTMI normalization. The cardinal temperature model with inflection is a three-parameter growth-rate law: zero below T_{min} , maximal at T_{opt} , zero at T_{max} , with a smooth asymmetric rise and a steep terminal fall - the asymmetry being the point, since microbial thermal death is much sharper than thermal limitation. We use the published form directly and normalize by its value at the 30 °C reference, so f_T is a pure capacity factor centered at one in the reference environment and the reference environment is exactly the base model. The cardinal values are anchored to published *S. cerevisiae* fits rather than fitted by us; the one we treat as fragile, $T_{max} = 42$ °C, is swept to 40 and 44 in the locked uncertainty arm, and Section 4.7 shows the G2 statistic's fate rides on that sweep. We regard that transparency - the fragile parameter is named, swept, and allowed to break a gate - as the correct handling of a parameter the literature has not pinned.

Levenspiel anchoring. The product-inhibition factor $(1 - P/P_{max})^n$ is the classical generalization of Monod kinetics to product inhibition, and its anchoring here uses published ethanol-inhibition kinetics for *S. cerevisiae*: biomass yield falling from 0.156 to 0.026 over 0–107 g/L ethanol, which the form reproduces qualitatively with P_{max} at the locked 14% v/v. The exponent $n = 1.5$ is marked ASSUMED: the literature constrains the endpoints of the curve, not its curvature, and the $\pm 50\%$ sweep (0.75–2.25) covers concave-through-convex shapes. The locked decision to apply f_E to the ethanol-exchange upper bound - rather than to the growth cap alone - is what makes product toxicity a fermentation- rate phenomenon in the model, and it is declared as a modeling choice with its alternative noted.

Maintenance form. Classical chemostat physiology shows maintenance energy demand rising as conditions deteriorate, but no consensus functional form exists for stress-dependent maintenance in this organism. We chose the reciprocal form $0.7 / \max(\min(f, 1), 0.01)$ because it is the simplest form with the required signs - flat and equal to the base NGAM (0.7 mmol ATP/gDW/h) when unstressed, rising without bound as the combined capacity factor approaches the viability floor - and we marked it ASSUMED in Amendment 1 with the floor $f_{floor} = 0.01$ declared. The viability

boundary itself (environments with $f < 0.01$ classified nonviable rather than collapsed) exists so that “the cell is dead” and “the cell is alive but unproductive” are different rows in every table; 73 of the 369 re-verified collapsed environments are of the former kind.

Glycerol physiology. Osmoadaptation in yeast is glycerol: the HOG pathway induces the GPD1/GPP1 branch, glycerol accumulates intracellularly to balance external water activity, and its biosynthesis diverts carbon from the glycolytic mid-point. The locked form - an obligatory export lower bound linear in external osmolarity, $k_{\text{gly}} = 1.0 \text{ mmol/gDW/h per M}$ - is the cheapest parameterization of that carbon invoice. The coefficient is ASSUMED (swept 0.5–2.0) because published glycerol yields under osmotic stress span a wide range across strain and regime; what the literature does fix is the sign and the approximate magnitude of the tax, and the A4 result (drain binding in 69% of solvable collapsed environments) is robust across the sweep. We note the deliberate absence of a HOG regulatory layer: the model assumes the adaptation happens and charges for it; it does not model the signaling.

Medium and objective. Two base-model choices carry more weight than their one-line declarations suggest. The rich complete-Y7 medium is the industrial-faithful choice - molasses and cane-juice mashes are not minimal media - and it is also the root cause of the inert nitrogen lever, since rich media feed nitrogen through amino acids; the paper’s largest limitation is the price of the same choice that makes the carbon economics realistic. The objective throughout is maximal ethanol exchange, never growth: growth appears only as the joint cap and through maintenance. This is the industrial objective (a producer buys titer, not biomass), and it is also what makes the design campaign’s question sharp, because growth-coupled design heuristics do not apply and every candidate is scored on the product objective directly.

3.5 Environment space, sampling, and the locked split

The environment space is the Cartesian product of four locked axes: temperature 28–42 °C (15 levels), ethanol 0–14% v/v (15 levels), osmolarity 0–1.5 M sorbitol-equivalent (16 levels), and nitrogen 5–100% of standard (20 levels): 72,000 environments. Single-axis and pairwise maps use the full grid; the combined-stress collapse surface uses a Latin-hypercube sample of 5,000 four-axis combinations (seed 20260926). Before any optimization, the sampled environments were split 70/30 into DESIGN and TEST regions by a stratified seeded split (seed 424242) committed in code; all search, tuning, and exploration used DESIGN only, and TEST was touched exactly once, at the final claim stage of the strain-design campaign. Two smaller locked subsamples serve the design campaign’s staging: a 40-environment screen sample (seed 557) for high-throughput screens and a 150-environment confirm sample (seed 558) for candidate confirmation, with a 150-environment test sample (seed 559) for the final verdict.

3.6 Mapping environments to constraints: implementation contract

The four stress layers are applied to the model in a fixed order on a fresh copy of the base reconstruction per environment, and the order and freshness are part of the method, not an implementation detail. Freshness guarantees that no state leaks between environments - each solve sees exactly

the pinned SBML plus one environment’s worth of constraint changes - which is what makes the per-environment rows of every artifact independent records, re-solvable in isolation. The application order (temperature, ethanol, osmotic, nitrogen, then the joint growth cap and maintenance computed from the combined factor) fixes the semantics of interaction: interactions in this model arise from the arithmetic of shared capacity factors and from the solver’s allocation of a shrinking carbon budget, never from hidden cross-terms in the code. When Section 4.7 reports sub-additivity, it is therefore reporting a property of the declared forms and the network, not of an interaction term we inserted - there is none to insert. Every layer application is logged with the parameter set hash, so the sweep arm (which changes parameter values) and the nominal arm differ only in their parameter records, and a reader can diff them.

The sampling design follows the same contract. Latin-hypercube sampling over the four axes (seed 20260926) gives uniform marginal coverage with far fewer points than the full grid - 5,000 against 72,000 - and the seeded DESIGN/TEST split (70/30, seed 424242) stratifies on collapse status so neither region is accidentally enriched in easy or hostile environments. The three nested subsamples (screen 40, design 150, test 150; seeds 557–559) exist for one reason: to bound how often the design campaign’s selection step could see the environments its final verdict would be scored on. The numbers are locked and small by design; we would rather defend a 40-environment screen than explain why a larger one was safe.

3.7 Engineering for auditability

Because the campaign ran on commodity sandbox machines subject to interruption, the compute layer was engineered around one requirement: no machine event may be able to silently change a result. Concretely: models are loaded fresh per evaluation; environments are evaluated in shards that checkpoint to disk (the A2 shard files are in the repository); long stages write progress journals (`test_progress.jsonl` and its siblings) so a resumed stage re-evaluates only unfinished designs; the evolutionary runner rebuilds its breeding population deterministically from the previous generation’s recorded scores plus the seed stream, so a mid-generation restart cannot change what gets bred; and every summary artifact carries the SBML hash and the pre-registration hash it answers to. The two bug classes this caught are on the record: a subsample-index mapping error caught by A4’s mandatory collapse-flag re-verification before any results were kept, and two list-hashability defects in the evolutionary resume path, fixed pre-outcome and recorded. The sixteen solver time-outs of Appendix C are the residual irregularity the engineering could not prevent, only bound - hence the register rather than a retry policy.

3.8 Objectives

The four locked objectives are the industrial ones: (O1) ethanol yield - the maximal ethanol-exchange flux relative to the same-environment wild type; (O2) temperature tolerance - the highest temperature at which a reference yield fraction remains feasible; (O3) ethanol tolerance - the highest ethanol concentration likewise; (O4) low nutrient requirement - the lowest nitrogen fraction likewise. Growth is never an objective; it appears only as a feasibility constraint. Tolerance metrics are defined relative to locked yield thresholds, and we warn in the methods that on this model class the tolerance readouts are mapping-limited: ethanol tolerance is governed by the f_E crossing of the Levenspiel cap, temperature tolerance by the CTMI maximum, and nitrogen tolerance by

the (inert) ammonium lever. They are reported as declared consequences of the mapping, never as biological discoveries; the objectives that carry the paper’s claims are O1 and the constraint-level analyses built on it.

3.9 Analysis stages

The program is staged A1–A8, each stage pre-registered, each artifact checksummed, and each gate reported as landed. A1: single-stressor dose responses on the full locked grid. A2: the combined-stress collapse surface on the 5,000-environment sample, with collapse defined as ethanol yield below 20% of reference. A3: non-additivity - the fraction of collapsed environments whose combined yield falls below the product of the single-stressor yields, with bootstrap confidence intervals and, per amendment, one-at-a-time parameter sweeps ($7 \text{ sets} \times 1,000 \text{ environments}$). A4: bottleneck attribution on the collapsed set - every collapse flag re-verified by fresh solve (0 mismatches), binding constraints counted with shadow prices, and flux-variability analysis on a locked 121-reaction subset over 25 solvable seed-777 environments. A5: strain design under Amendment 4 - all 1,143 single-gene deletions; a seeded double-knockout screen (435 designs); 36 reaction-bound scalings ($0.25 \times -4 \times$ on a locked reaction list); an evolutionary search ($3 \text{ seeds} \times 12 \text{ generations} \times \text{population } 24$, genome = top-30 confirmed modifications); a matched random baseline (150 random budget-2 designs, 30 seeds); staged screen/confirm/test evaluation; and locked plausibility filters (essentiality under the target-regime exemplar; dataset concordance pending at design time). A6: validation against the locked deletion-screen datasets (Section 3.14) with locked overlap statistics. A7: regime–transcriptome concordance against the Gasch-2000 environmental-stress expression program over locked marker gene sets (heat-shock, nitrogen-catabolite, glycerol, glycolytic), with SGD-verified gene identifiers frozen in code. A8: dynamic heat-wave simulations (dynamic FBA over a locked time grid), with the batch substrate-exhaustion confound declared and the very-high-gravity variant labeled exploratory (post-run, Section-12 rule).

3.10 Stage protocols in detail

The stage summaries above are the contract; the protocols below are the execution. Every stage runs as an independent driver with its own artifact directory, writes a summary JSON carrying the pre-registration hash it answers to, and can be resumed after machine interruption without recomputation.

A1 protocol. For each single axis, the environment is varied over its locked levels with all other axes at reference. At each point the stress layers are applied to a fresh copy of the model (models are never mutated cumulatively), feasibility is checked, and maximal ethanol exchange is solved. The relative yield is the ratio to the same-day reference solve. The 66-row artifact (`a1_dose_response.csv`) records the axis, level, absolute yield, feasibility, the SBML hash, and the relative yield, so the dose-response figure is a join, not a reconstruction.

A2 protocol. The 5,000 four-axis environments are drawn by Latin-hypercube sampling over the locked ranges (seed 20260926), partitioned into 200-environment shards, and each shard is solved independently with shard-level checkpointing - the shard files are part of the repository. An

environment is collapsed when its relative ethanol yield falls below 0.20; the flag, the yield, and the solver status are recorded per environment. The collapse surface is then a pure query over the joined artifact, which is what makes the A4 re-verification (below) meaningful.

A3 protocol. For every sampled environment, the three single-stressor yields at its axis values are looked up from the A1 grid (bilinear interpolation is prohibited; environments off the A1 knots use their own single-axis solves), the Bliss baseline is the product, and the deviation is the observed combined yield minus the baseline. The locked statistic is the fraction of collapsed environments with negative deviation, with a bootstrap CI over environments (seed 20260927). The Amendment-2 sweep arm re-runs the statistic on 1,000-environment samples under each one-at-a-time parameter perturbation - seven sets total - so the gate decision sees the statistic's parameter sensitivity, not a point estimate.

A4 protocol. The collapsed set is subsampled deterministically (every fifth environment, 369 of the sample), and every subsampled flag is re-verified by a fresh solve before any analysis - this re-verification caught and quarantined an indexing bug in an early run, before any results were kept. For each re-verified collapsed environment, the solver's bound activity at the optimum is read directly: a constraint binds when its flux equals its bound within tolerance, and its shadow price (reduced cost) is recorded from the dual. Binding-set clusters are counted over environments. The FVA arm fixes the objective at its optimum and varies each member of the locked 121-reaction subset over 50 seeded environments (seed 777), of which 25 were solvable; both the strict forced-width criterion and the direction-aware check (does the optimum sit at the bound for lower-bound constraints?) were locked in advance, and both are reported, including where they disagree.

A5 protocol. The design campaign is staged to protect the TEST region: screens ran on the 40-environment screen sample, the top-30 per arm confirmed on the 150-environment design sample, and only confirmed candidates (25) touched the 150-environment TEST sample, exactly once. Singles: all 1,143 genes with model reactions, deletion by GPR knock-out. Doubles: 435 seeded pairs from the locked candidate pool. Scalings: 36 reaction-bound multipliers ($0.25\times$, $0.5\times$, $2\times$, $4\times$ on a locked nine-reaction list covering glycerol, glycolytic, fermentative, and membrane-transport nodes). Evolutionary search: three seeds, twelve generations, population 24, genome over the top-30 confirmed modifications, selection on mean DESIGN-relative yield with collapse penalty. The random baseline draws 150 budget-2 designs (30 seeds) under identical evaluation, giving the null distribution G4(b) compares against. Plausibility filters were locked before outcomes: filter (a) excludes designs essential under the target-regime exemplar; filter (b) - dataset concordance - was pending at design time by construction (the datasets are A6's subject) and is recorded as such. Every evaluation ran under a hard per-design timeout with the solver-timeout register of Appendix C.

A6 protocol. For each dataset, the locked universe is the intersection of model genes and measured genes; the model side is the locked stress-conditional sensitivity call of Section 3.14 (growth-based, exclusion rule for unconditionally sick knockouts recorded per gene); the dataset side is the locked hit call. The overlap is a Fisher exact test (one-sided greater) per comparison with BH-FDR 5% across the four comparisons, plus the MCC with bootstrap CI. The qualitative arm froze

the named genes of the AJEV screen in code with SGD-verified systematic identifiers before any model call; unresolved identifiers (TDP3) were recorded as unresolved, not substituted.

A7 protocol. Marker gene sets for the four regimes were locked with citations (glycerol biosynthesis; heat-shock proteins; glycolysis; nitrogen-catabolite targets), systematic identifiers SGD-verified and frozen in code, and each marker's log2 ratio read from the locked Gasch-2000 column for its regime's shock. Agreement is directional (induction where the regime is active), scored by one-sided binomial test per arm. The ethanol arm was locked as untestable - Gasch-2000 contains no ethanol-stress experiment - and was not retrofitted.

A8 protocol. Dynamic FBA over the locked time grid (0.25 h steps, 16 h), with the environment (temperature) updated per phase of the locked heat-wave profile, batch mass balances integrated explicitly (biomass, glucose, ethanol, glycerol; 1 L, 0.1 gDW/L inoculum, 111 mmol/L initial glucose), and declared unit constants. Dynamic collapse onset is the first step at which the relative ethanol flux falls below the collapse threshold; the static prediction is the same threshold applied to the static surface along the trajectory. The batch substrate-exhaustion confound was declared in the amendment, and the very-high-gravity variant (1,665 mmol/L) is labeled exploratory per the post-run rule.

3.11 Objective definitions and tolerance metrics

The four locked objectives share one measurement discipline: every quantity is evaluated per environment against a same-environment wild-type reference solved in the same session, so no comparison ever crosses a solver, a parameter set, or a day.

O1, ethanol yield, is the maximal ethanol-exchange flux under the environment's full constraint set, divided by the wild type's under the same constraints. Collapse is the locked 0.20 relative-yield threshold. We chose relative rather than absolute yield because absolute flux scales with the medium's carbon supply, and relative yield isolates the stress effect; the cost of the choice is that relative yield saturates at 1.0 for any modification whose effect is already realized by the wild type - which is precisely the measurement regime A5 found itself in, and one reason the null is informative rather than an artifact of scale.

O2 and O3, temperature and ethanol tolerance, are threshold readouts: the highest axis value at which a locked reference yield fraction remains feasible, scanned on the single-axis grid. They are declared mapping artifacts wherever reported (the Levenspiel crossing for ethanol, the CTMI maximum for temperature), and their role in the gates is limited to G4's dominance test, where identical-to-WT tolerances are the expected outcome under a null - which is what was observed, 25/25.

O4, low nutrient requirement, is the lowest nitrogen fraction at which the locked yield fraction remains feasible. It saturates at the 5% grid edge for every design and the wild type alike - the flat-axis artifact, root-caused in A6 to the inert ammonium lever and declared wherever the objective appears.

The dominance test of G4(a) requires a candidate to match or exceed the wild type on all four objectives with strict improvement on at least one, beyond the declared 10^{-14} solver-noise band. The band exists because COBRA solutions carry floating-point path dependence at the 10^{-15} scale,

and counting an 8×10^{-15} difference as a biological improvement would be the exact failure mode pre-registration exists to prevent.

3.12 Dataset acquisition and processing

All validation data are public, pre-existing, and acquired at accession level with SHA-256 hashes recorded in `data/raw/SECTION8_MANIFEST.json`; no dataset was filtered, normalized, or subset after outcomes were visible.

GSE151784 was pulled from GEO as raw barcode-count tables (pooled deletion library, before/after acute lethal ethanol, triplicate), parsed per ORF, averaged across replicates on the $\log_2$ depletion scale, and thresholded at the locked -1 hit call; replicate agreement was inspected before locking the call and is recorded in the processing script.

Gibney 2013 (PNAS, doi:10.1073/pnas.1318100110) was obtained as the published supplementary Bar-seq tables; the publisher’s site refuses automated access, so the supplements were mirrored through the PubMed Central instance’s binary endpoints, with the access path recorded. The authors’ own per-gene significance calls were used as the locked hit definition - no re-derivation.

GSE59659 was taken as the processed series matrix (per-ORF log-ratios for the 20%-glucose and 1 M-sorbitol arms) mapped to systematic identifiers through the GPL9825 platform table; per-ORF medians across replicates were thresholded at the locked -1 call.

AJEV 59(4):401 is paywalled; the named genes were taken from the abstract and public indexing, verified against SGD, and frozen in code with their systematic identifiers before any model evaluation. One name (TDP3) could not be resolved through the SGD API and is recorded as unresolved; the paper’s attribution note (Renuka Kumar et al., Bisson laboratory, not van Voorst) is on record in the A6 run log.

Gasch 2000 was recovered as the authors’ complete dataset file from a Wayback Machine capture of the Stanford site (SHA-256 recorded), used only for A7 concordance, and graded context by the pre-registration - never validation.

E-GEOD-1723 and *E-MTAB-8245* were acquired for the nutrient checks and became not evaluable with the inert-lever finding; their accession records stand in the manifest.

3.13 Gates

Four binary gates were locked before outcomes. G1 (model validity): wild-type dose responses qualitatively reproduce known stress physiology (monotonic decline per axis; heat-ethanol interaction consistent with the industrial literature) AND predicted stress-conditional sensitivity significantly overlaps deletion-screen hits (Fisher odds ratio > 1 , BH-FDR 5%) in at least 2 of the 4 quantitative comparisons. G2 (non-additivity): the sub-additive fraction exceeds 25% of collapsed environments with bootstrap CI excluding zero. G3 (bottlenecks): collapse is attributable to named constraints with consistent shadow-price and variability evidence. G4 (strain design): at least one candidate dominates the wild type on all four objectives with strict improvement on at least one (G4a) and exceeds the 95th percentile of the matched random baseline on ethanol yield (G4b), under the plausibility filters. The gates are reported in Section 4 exactly as they landed: G1 partial (dose-response clause met; overlap clause failed 0/4), G2 not met (24.2%, parameter-fragile), G3 met, G4 not met (0/25 beyond solver noise).

3.14 Validation datasets and statistics

Validation uses existing experimental data only, at accession level, with hashes recorded in the dataset manifest and no parameter fitted to any validation outcome. Quantitative: GSE151784 (pooled deletion-library barcode counts before/after acute lethal ethanol, triplicate), the Gibney-2013 Bar-seq heat-shock survival screen (PNAS, doi:10.1073/pnas.1318100110; supplements mirrored via PMC), and GSE59659 (deletion fitness under 20% glucose and 1 M sorbitol; processed series-matrix values with the GPL9825 probe map). The locked overlap metric is the stress-conditional sensitivity overlap: Fisher exact odds ratio (one-sided greater) with Benjamini–Hochberg FDR at 5% across the four comparisons, and the Matthews correlation coefficient with a 95% bootstrap CI (10,000 resamples, seed 20260927). Model-side sensitivity is locked as a growth-based call: for gene g , $\text{sens}(g) = [g_{\text{KO}}(\text{stress})/g_{\text{KO}}(\text{ref})]/[g_{\text{WT}}(\text{stress})/g_{\text{WT}}(\text{ref})]$, sensitive when $\text{sens}(g) < 0.8$; unconditionally sick or essential knockouts (reference growth $\leq 10^{-6}$) are excluded and recorded. Dataset-side hit calls are locked per dataset: GSE151784 mean $\log_2$ depletion < -1 across replicates; Gibney the authors’ own significance call; GSE59659 per-ORF median log-ratio < -1 averaged across replicates. Qualitative: the named genes of the AJEV 59(4):401 ethanol screen (concordance stated per gene), and the two nutrient datasets E-GEOD-1723 and E-MTAB-8245, which proved not evaluable because the nitrogen lever is inert (Section 3.3). The Gasch-2000 expression program is context for A7 and is never used as validation of flux predictions.

3.15 Pre-registration protocol and integrity register

The full design above - grid, forms, parameters, stages, gates, datasets, and statistics - was locked in docs/PREREGISTRATION.md with its SHA-256 recorded at lock time, before any outcome run. Later amendments (viability boundary and maintenance form; the parameter-sweep arm; the A5 staging and filters) were themselves committed before their first runs, with their own hashes. Solver irregularities are a registered category, not an inconvenience: GLPK can hang in native code on certain perturbed models, so every candidate evaluation runs under a hard per-design timeout and a hang is recorded as `solver_timeout` - 16 occurrences across the campaign - never silently skipped or imputed. All code, data, run logs, and the judge archive are in the repository; the paper’s every number can be regenerated from the pinned commit.

3.16 The pre-registration lifecycle, as executed

The discipline contract operated in four phases, and each left artifacts. *Lock*. The design document was written before any outcome run: environment grid and axes, stress-layer forms with parameter provenance, the four objectives and their thresholds, the eight analysis stages, the four gates with their binary criteria, the validation datasets with accession identifiers, and every statistical test with its direction, correction, and seed. The document’s SHA-256 was recorded at lock time and is embedded in every stage artifact that answers to it. *Amendment*. Four amendments were needed - the viability boundary and maintenance form (Amendment 1), the parameter-sweep arm (Amendment 2), the transcriptome and dynamic stages (Amendment 3), and the A5 staging and plausibility filters (Amendment 4) - and each was committed with its own hash before the first run it governed; no amendment touched a locked objective, gate threshold, grid, or statistic after outcomes existed. *Execution*. Stages ran in dependency order with per-stage artifacts, and

two categories of mid-flight correction were allowed and are on record: bug fixes committed pre-outcome (the A4 index-mapping fix, the evolutionary resume fixes, the A6 WT-evaluation fixes), and resolver-level engineering (checkpointing, timeouts) that cannot change an outcome’s definition. *Scoring*. Gates were scored against the locked criteria exactly as outcomes landed, and the exploratory label was applied per the post-run rule to everything outside the locked design - in this campaign, the very-high-gravity dynamic variant. The result is a paper in which the word “locked” is always checkable: every locked quantity names the document and hash that locks it.

3.17 Solvers, tolerances, and numerical hygiene

All optimization used GLPK through COBRAPy defaults with two declared exceptions: per-evaluation worker-termination timeouts (bounded, with the timeout register of Appendix C), and the declared 10^{-14} noise band for relative-yield comparisons. The band’s origin is empirical: repeated solves of identical models on identical data returned relative yields differing at the 10^{-15} scale across solver invocations, and the band was set an order of magnitude above the observed spread before any comparison was scored. Feasibility was taken from solver status, never from flux magnitude; the viability classification uses the declared $f_{\text{floor}} = 0.01$ on the combined capacity factor; and collapse uses the locked 0.20 relative-yield threshold. Flux-variability analysis fixed the objective at its optimum exactly (no optimality slack), so the forced-width criterion of A4 is conservative by construction - which is part of why the strict criterion failed and the directional criterion carries the weight it does. Every statistic that resamples (bootstrap CIs, the random baseline, the LHS, the splits) names its seed in the methods, and all seeds are single-use.

3.18 Statistical methods, collected

Every inferential statement in the paper uses one of five locked procedures, collected here so the results sections can simply cite them. (i) *Bootstrap over environments*: confidence intervals for fractions defined on the environment sample (the A3 negative-deviation fraction) are percentile bootstrap intervals over 10,000 resamples of environments, seed 20260927. (ii) *Fisher exact overlap*: dataset concordance in A6 is a one-sided (greater) Fisher exact test on the locked 2×2 table (model-sensitive or not $\times$ dataset hit or not) over the locked universe, with Benjamini–Hochberg correction at 5% across the four comparisons. (iii) *Matthews correlation*: effect size for the same tables is MCC with a 95% bootstrap CI (10,000 resamples, same seed); degenerate margins produce MCC 0 with CI $[0, 0]$, which we report rather than suppress. (iv) *Binomial concordance*: A7 arms are scored by one-sided binomial tests against chance directional agreement, with Clopper–Pearson intervals. (v) *Empirical baseline*: G4(b) is an empirical-percentile comparison against the 150-design random baseline, no distributional assumption. Thresholds (0.20 collapse, 0.8 sensitivity, $-1 \log_2$ depletion, 10^{-14} noise) are locked constants, never tuned; two of them (0.20, 0.8) are conventional, and the threats-to-validity section owns their sensitivity. No p-value anywhere in the paper is computed on data that informed its own hypothesis: the A6 universes, hit calls, and model calls were all locked before the first overlap was computed.

4 Results

All numbers below are regenerated from the artifacts at the pinned commit; nothing is quoted from memory. Gate outcomes are reported exactly as the pre-registration scored them, including the three outcomes that went against the program’s ambitions.

4.1 A1: single-stressor dose responses (wild type)

The wild-type model reproduces the qualitative shape of all three active single-stressor response laws on the locked grid (66 environments). Maximal ethanol exchange rises from 43.1 mmol/gDW/h at 28 °C to the reference 45.46 at the 30–33 °C optimum plateau, then falls monotonically through 34.4 at 38 °C to infeasibility at 42 °C - the CTMI cardinal structure operating as designed. The ethanol axis follows the Levenspiel law: relative yield falls to 0.895 at 1%, 0.515 at 5%, 0.213 at 9%, and 0 at 13%. The osmotic axis shows the monotonic carbon-cost decline the glycerol-drain term imposes. The nitrogen axis is flat: as established in Section 3.3, the ammonium lever is inert in the rich medium, and every nitrogen-frac row returns the reference yield. We treat the three active curves as the G1 dose-response clause (met) and the nitrogen curve as declared evidence of the limitation, not as a dose response.

4.2 A2: the combined-stress collapse surface

Over the 5,000-environment Latin-hypercube sample, 1,857 environments (37.1%) are collapsed under the locked definition (relative ethanol yield below 0.20). The collapse geography is structured, not diffuse: collapse concentrates along the high-temperature and high-ethanol edges and their joint corner, with the osmotic axis shifting boundaries rather than creating them, and the nitrogen axis contributing nothing (by the declared construction). For the bottleneck analysis of Section 4.10, the collapsed set was subsampled deterministically (every fifth environment of the sample, 369 environments) and every collapse flag was re-verified by a fresh solve with zero mismatches; an earlier indexing bug in this subsample mapping was caught by that re-verification and fixed before any results were kept.

4.3 A1 in detail: the shape of each axis

The temperature curve deserves a careful reading because it is the axis on which the G2 fragility later turns. From 28 °C (relative yield 0.948) the curve rises to its optimum plateau across 30–33 °C, where the CTMI normalization holds it at the reference value; the decline then steepens through 34 °C (0.996), 36 °C (0.914), and 38 °C (0.757), passes the half-yield point just above 40 °C (0.502 at 40 °C exactly), and reaches infeasibility at 42 °C. Two properties of this shape matter downstream. First, the plateau means that in the mild corner of the grid, temperature is a non-stressor: nearly a third of the sampled environments sit on the plateau side of the surface. Second, the steep terminal fall means that the placement of $T_{\max}$ - the one cardinal parameter without a tight published consensus - moves the collapse boundary by whole percentage points of ethanol tolerance, which is precisely the mechanism by which the $T_{\max}$ sweeps break the G2 statistic (Section 4.7). The fragility is not hidden in the fit; it is the physics of the cardinal form near its maximum.

The ethanol curve is the Levenspiel law operating as declared: relative yield 0.895 at 1%, 0.794 at 2%, 0.604 at 4%, 0.515 at 5%, 0.354 at 7%, 0.281 at 8%, 0.153 at 10%, 0.054 at 12%, and infeasibility at 13% - one percentage point below the locked $P_{\max} = 14\%$, the gap being the joint effect of the growth-cap and maintenance terms as $f_E \rightarrow 0$. We flag the obvious consequence: because the 20%-collapse threshold is crossed at about 5.4% ethanol on this curve, a large fraction of the A2 collapse set is ethanol-driven, and the collapse map's ethanol edge is a mapping edge. What the map adds beyond the single-axis law is the interaction structure of Section 4.6, not the edge itself.

The osmotic curve is shallow by design of the locked coefficient: relative yield declines linearly from 1.000 at 0 M through 0.989 at 0.5 M to 0.968 at 1.0 M and 0.957 at 1.5 M - a 4.3% carbon tax at the grid's osmotic maximum, far above the 20% collapse threshold on its own. The osmotic axis therefore cannot create collapse in isolation under the locked k_{gly} , and its role in the combined surface is as a boundary-shifter: it subtracts from the carbon budget of environments already near the edge. The nitrogen curve is flat at 1.000 across all twenty levels, which is the visible signature of the inert-lever limitation declared in Section 3.3; we keep the twenty rows in the artifact because the flatness is evidence.

4.4 A1 numbers walk: the three active curves and the flat fourth

The temperature curve's full sixteen points, because they carry the A3 fragility: 28 °C 0.948, 29 0.977, 30–33 the 1.000 plateau, 34 0.996, 35 0.963, 36 0.914, 37 0.846, 38 0.757, 39 0.643, 40 0.502, 41 0.329, 42 infeasible. Note the plateau's width - four grid points - and the fall's acceleration: the curve loses the last half of its value in two degrees. The ethanol curve's thirteen points: 0% 1.000, 1 0.895, 2 0.794, 3 0.696, 4 0.604, 5 0.515, 6 0.432, 7 0.354, 8 0.281, 9 0.213, 10 0.153, 11 0.099, 12 0.054, 13 infeasible; the 0.20 collapse threshold sits between 8% and 9% on this axis alone. The osmotic curve declines at 0.0022 relative yield per 0.1 M across its sixteen levels, reaching 0.957 at 1.5 M - never collapse-capable in isolation, always a budget subtraction. The nitrogen curve is 1.000 at all twenty levels, the flatness that A6 root-caused. These 66 solves are the cheapest in the campaign and the most quoted: every relative number downstream divides by a curve defined here, which is why A1 runs first in the dependency chain and why its artifact carries the reference hash every later stage cites.

4.5 The DESIGN/TEST discipline, quantified

The split protocol's guarantees are worth stating as numbers. The DESIGN region (70% of each locked sample) absorbed: all 1,143 single screens, all 435 double screens, all 36 scaling screens, all 864 evolutionary evaluations (3 seeds $\times$ 12 generations $\times$ 24), all 150 baseline designs across 30 seeds, and every confirm-stage evaluation (90 candidates across four arms) - on the order of four thousand model evaluations across the campaign. The TEST region absorbed exactly 25 evaluations, once each, at the final stage, plus the wild-type reference needed to score them. No threshold, filter, or selection rule was adjusted after a TEST result existed; the one rule that touched TEST outcomes - the noise-band discount on the YGL248W literal pass - was locked in the methods before the TEST run. The asymmetry is the point: the campaign was free to explore as aggressively as its compute allowed precisely because the verdict stage was rationed. We recommend

the pattern to any in silico design study: the budget that matters is not solver time, it is how many times the held-out region is touched, and the honest answer to that question here is once.

4.6 A2 in detail: the geography of collapse

Marginal collapse frequencies across the 5,000-environment sample make the interaction structure quantitative. Temperature: 44.3% of environments at or above 38 °C are collapsed, against 32.1% at or below 32 °C. Ethanol: 100% of environments at or above 10% v/v are collapsed - the mapping edge - against 3.8% at or below 2%. Osmolarity: 38.1% at or above 1.2 M against 34.5% at or below 0.2 M, a modest but real shift consistent with the shallow single-axis tax. The mild corner (temperature ≤ 32 °C, ethanol $\leq 2\%$, 191 sampled environments) contains zero collapses; the hostile corner (temperature ≥ 38 °C, ethanol $\geq 10\%$, 415 environments) is entirely collapsed. Between the corners, the surface is a smooth ramp whose gradient is set jointly by the thermal and ethanol terms, with the osmotic term translating it.

Read as an operational map, the surface says: there is no free headline temperature gain to be had against ethanol, and no free ethanol gain against heat - the joint region where a producer would want to run hot (to save cooling) and push titer (to save distillation energy) is exactly the region where the model's carbon economy closes. This is the map the design campaign then tried, and failed, to move.

4.7 A3: non-additivity is real but not parameter-robust

Under the locked Bliss-independence definition, 450 of the 1,857 collapsed environments (24.2%; bootstrap 95% CI [22.3%, 26.1%]) show combined yield below the product of their single-stressor yields. The primary estimate sits just under the locked 25% gate, and the pre-registered one-at-a-time parameter sweeps (7 sets, 1,000 environments each) show the fraction ranging from 12.8% to 33.5%: it meets the gate at nominal parameters and under both glycerol-cost and both maintenance-exponent perturbations, but fails under both temperature-margin perturbations ($T_{\max} = 40$: 18.7%; $T_{\max} = 44$: 12.8%). Per the locked rule, G2 is therefore **not met**: a negative-deviation excess in collapsed environments exists, but its magnitude is set by the thermal-death shape parameter, so it is not a parameter-robust law of the model. We note for the record that the mean deviation in collapsed environments is small in absolute terms (0.2% of reference); the negative tail is concentrated, not pervasive.

4.8 A3 in detail: reading the sweep table

The parameter-uncertainty arm is the honest core of the non-additivity analysis, and its table rewards a slow reading. At nominal parameters, the 1,000-environment resample gives 369 collapsed environments with a negative-deviation fraction of 26.3% - above the locked 25% gate, as are both glycerol-cost perturbations (27.2% and 25.9%) and both maintenance-exponent perturbations (33.5% and 27.3%). Under $T_{\max} = 40$ °C the fraction falls to 18.7%, and under $T_{\max} = 44$ °C to 12.8%; the primary 5,000-environment run at nominal parameters gives 24.2% (CI [22.3%, 26.1%]), a hair under the gate. The pattern across the seven sets is not noise around a stable mean - it is a monotone sensitivity to one parameter family, the thermal-death margin, with the collapse fraction itself swinging from 16.7% to 50.1% of sampled environments as the temperature margin

moves. The interpretation we print is the one the table forces: a negative-deviation excess in collapsed environments is a real feature of this model's joint response surface - it appears in every set - but its magnitude relative to any fixed threshold is set by the thermal-death shape, so the gate, which was locked as a magnitude claim, fails. A weaker gate ("is the fraction ever above zero?") would pass trivially and prove nothing; a stronger one ("is it above 25% robustly?") fails informatively. We also record, against over-reading, that the mean signed deviation across collapsed environments is positive and small in every set (0.09–0.96% of reference): the sub-additivity is a concentrated tail, not a pervasive synergy, and the paper makes no claim of general stress synergy anywhere.

4.9 Stage interlocks: how the results depend on each other

The stages were designed as a dependency chain, and the chain's structure is part of the result. A1 calibrates nothing but defines everything: the single-axis laws are the denominators of every relative quantity in the paper. A2 consumes A1's grid to locate collapse; A3 consumes both to score deviation from independence; A4 consumes A2's collapse flags - re-verified, with the indexing bug that re-verification caught on record - and explains them; A5's search space and its null only make sense given A4's mechanism, and A6's degenerate overlap tables are the same mechanism seen from the gene side; A7 independently confirms the regimes A4 attributes; A8 lifts the static surface into time. Two consequences follow for readers re-using parts of this work. First, the artifacts compose: any stage can be re-run against the pinned model and the upstream artifacts without rerunning the campaign, and every summary JSON names the pre-registration hash it answers to. Second, the failure modes also compose: the inert nitrogen lever propagates flatness into every N-indexed row downstream, and it is labeled at each stage rather than corrected once, because a single silent correction would have made the propagation invisible.

4.10 A4: what actually binds when productivity collapses

Of the 369 re-verified collapsed environments, 199 solve to an optimal but low-yield state, 97 are infeasible, and 73 sit below the viability boundary. Across the 199 solvable collapsed environments, the binding constraint counts are: environment-imposed ethanol cap 197, osmotic glycerol drain 138 (69% of the solvable set), glucose-uptake cap 118 (59%), and joint growth cap 39. The two dominant clusters are ethanol-cap + glycerol-drain (66 environments) and glucose-cap + ethanol-cap + glycerol-drain (49). Dual evidence agrees: when the glucose cap binds, its mean shadow price is 0.58 units of ethanol per unit of uptake capacity; when the glycerol drain binds, 0.37 per unit drain; the growth cap, where it binds, is degenerate (zero shadow price). Flux-variability analysis on the locked 121-reaction subset over 25 solved seed-777 environments is directionally consistent in 25/25 cases: the optimum sits at the glucose-uptake bound and at the glycerol-drain bound simultaneously, so neither carbon supply nor the osmotic carbon tax can be relaxed without immediate objective gain.

The substantive finding of the paper is this co-limitation: in combined-stress collapse, carbon supply (thermally capped uptake) and an obligatory carbon tax (osmoregulatory glycerol) bind together and dominate the model-imposed ethanol-inhibition cap as intervention targets. Two caveats are part of the result, not footnotes to it. First, under the strict pre-declared forced-width FVA criterion only the ethanol-cap cluster passes, and that cap is the objective carrier by construction, so

it cannot count as an independent discovery; G3 is scored **met in substance** on the binding-plus-dual-plus directional-FVA evidence for the glycerol and glucose clusters, with the strict criterion declared failed. Second, the ethanol cap's near-universal binding (197/199) is a consequence of how the objective is posed, and is never cited as a finding.

4.11 A5: strain design finds nothing (an exhaustive honest negative)

The strain-design campaign asked the strongest question the model supports: does any network modification - deletion, flux scaling, or an evolved combination - dominate the wild type on all four locked objectives? The search was deliberately broad: all 1,143 single-gene deletions; a seeded screen of 435 double knockouts; 36 reaction-bound scalings from $0.25\times$ to $4\times$ on a locked reaction list; and an evolutionary search (3 seeds, 12 generations, population 24, genome drawn from the top 30 confirmed modifications). Every screen, confirmation, and test ran on the locked environment subsamples, with the 150-environment TEST region touched exactly once.

The answer is no, at every level of the hierarchy. All 1,143 singles, all 435 doubles, all 36 scalings, and every evolved finalist return relative yield 1.0 against the wild type on the confirmed environments. On the final TEST evaluation, one candidate of 25 (a YGL248W knockout) prints a literal pass at relative yield $1 + 8 \times 10^{-15}$ - inside the declared 10^{-14} solver-noise band - with tolerances identical to the wild type (37 °C, 1% ethanol, 5% nitrogen). G4 is therefore scored **failed**: no candidate improves any objective beyond numerical noise, and the literal 1/25 count is noise, not a design.

The negative is mechanistically explained by A4, and the explanation is the point: collapse in this regime is driven by environment-imposed caps - the ethanol cap, the glycerol drain, the glucose-uptake cap - acting on the network boundary, so removing or rescaling internal network reactions cannot recover yield. Flux-balance strain design can only find gains that live inside the network; the binding constraints here live at its edge. A matched random baseline (150 budget-2 designs, 30 seeds) confirms the search was not underpowered: the confirmed candidates sit within the baseline distribution, which itself clusters at 1.0. Sixteen solver timeouts (GLPK native hangs on specific perturbed models, e.g. YCL052C deletion and several GPP/PDC/MEP scalings) were recorded in the register and never imputed.

4.12 A6: independent validation is an honest null

The validation campaign confronted the model with three quantitative deletion-screen datasets acquired at accession level with recorded hashes (Section 3.14), under the locked overlap statistics, plus the locked qualitative checks. The quantitative result is a clean null: across all four comparisons - ethanol (GSE151784: universe 705, 604 dataset hits), heat (Gibney 2013 Bar-seq: universe 678, 536 hits), and both osmotic arms (GSE59659: universe 1,060, 141 and 146 hits) - the model's stress-conditional sensitive set contains 0 genes in the ethanol and heat universes and 3 in the osmotic, and the overlap with the experimental hit sets is 0 in every comparison. The G1 overlap clause fails 0/4 at BH-FDR 5%, and is reported as such.

The qualitative ethanol check (the five evaluable named genes of the AJEV 59(4):401 screen, with SGD-verified identifiers frozen in code) concords in 0 of 1 evaluable cases: ADH1 deletion grows at wild-type rate in the model under 8% ethanol (isozyme redundancy is real in the network), while the screen reports sensitivity; GSH1 is excluded under the locked rule (unconditionally lethal

in the model, a declared artifact of absent glutathione salvage); CLC1, UME6, and VPS24 have no metabolic-model counterpart; TDP3 could not be resolved via SGD. The two nutrient checks (E-GEOD-1723 direction, E-MTAB-8245 spare capacity) are not evaluable, for the root cause declared in Section 3.3: the nitrogen lever is inert, so no nitrogen-limited state exists to check against.

We state plainly what this means. The deletion-screen biology of stress survival - protein synthesis, vacuolar function, mitochondrial maintenance, membrane trafficking - lives outside what flux balance with environment-imposed caps can see. The model is a valid map of carbon-flow economics under stress (Sections A1–A4) and an invalid predictor of gene-level stress sensitivity (A6). Both statements are results of this paper.

4.13 A5 in detail: anatomy of a null

The null deserves its own inspection, because “nothing works” can mean a weak search, a broken evaluation, or a true boundary, and only the third is worth publishing. Three internal checks say the third.

First, the evaluation could see effects where effects exist. The matched random baseline - 150 budget-2 designs drawn under identical code - does not sit uniformly at 1.0: its median mean-relative yield is 1.0, but 44% of random designs score below it, some substantially (the weakest random pairs land near 0.97 with up to 92 newly collapsed environments across the 150-environment design sample; the mean random design newly collapses about one environment). The measurement pipeline therefore resolves both damage and neutrality at the per-environment level; what it never sees, from any arm, is gain.

Second, the search was broad in kind, not only in count. Deletions remove function; scalings move capacity in both directions ($0.25\times$ to $4\times$) on a reaction list chosen to cover the glycerol branch, glycolysis, fermentation, and membrane transport - the very nodes the A4 mechanism implicates; the evolutionary search recombined the best of all of it for twelve generations under three seeds. If a network-internal lever existed anywhere near the confirmed modification set, the evolutionary stage - which starts from it - is the stage that should have found it. It found exactly the same 1.0.

Third, the explanation pre-dates the result. A4’s mechanism says collapse is enforced at the boundary - uptake capacity, product cap, obligatory glycerol drain - and boundary constraints in FBA are properties of the environment mapping, not of the network, so no GPR rule can delete them and no internal scaling can relax them. The A5 null is what A4 predicts. When a search outcome is the deductive consequence of an independently gated finding, the negative is a measurement, not an absence.

The one literal pass in the TEST evaluation deserves its sentence of record: YGL248W deletion printed $1 + 8 \times 10^{-15}$ on mean relative yield, a number eight orders of magnitude inside any conceivable biological effect and inside the declared solver-noise band, with wild-type-identical tolerances. We report it because the locked rules require counting it, and we score G4 failed because the same locked rules require passing to mean something.

4.14 A2 numbers walk: from 5,000 environments to a surface

The collapse surface is built from exactly 5,000 solves, and its construction is fully enumerable. Each of the 25 shards of 200 environments was solved independently; the joined artifact carries, per environment, the four axis values, the absolute maximal ethanol exchange, the feasibility flag, the relative yield, the collapse flag, and the SBML hash - no derived quantity is stored without its inputs. Of the 5,000 solves, 1,857 returned relative yield below the locked 0.20 threshold. The complement - 3,143 productive environments - is itself structured: productivity above 90% of reference occupies the mild corner and a wedge extending to moderate ethanol at low temperature, and the 0.5–0.9 band forms the transition ramp the interaction analysis of A3 then interrogates. The surface's three boundary types - smooth ramp along each single axis, knee where the thermal and ethanol ramps meet, and a hard infeasibility wall at the joint extreme - are the geometric content of the co-limitation mechanism before any dual analysis: two independent carbon gates produce a surface with a knee, and a single gate would have produced a smooth diagonal.

The deterministic subsample used downstream (every fifth environment, 369 collapsed of 1,000 sampled at nominal parameters) preserves this structure by construction, and the A4 re-verification - every subsampled flag re-solved from scratch, 0 mismatches - means the bottleneck counts inherit no stale-solver state. We note the one incident in this pipeline because the paper's discipline requires it: an early version of the subsample mapping misaligned environment indices between shards; the mandatory re-verification pass caught it before any analysis ran on the misaligned set, the fix was committed pre-outcome, and the incident is recorded in the A4 run log. This is what the re-verification step is for, and it is why it is mandatory rather than optional.

4.15 A4 numbers walk: binding counts and their denominators

The bottleneck counts are quoted with three different denominators in the literature of this kind of analysis, and we fix ours explicitly. Over all 369 re-verified collapsed environments: ethanol cap binds 197 (53%), glycerol drain 138 (37%), glucose cap 118 (32%), growth cap 39 (11%). Over the 199 environments that solve to an optimum (the denominator that matches the intervention question - what would relaxing this constraint buy?), the same counts read 197 (99%), 138 (69%), 118 (59%), 39 (20%). The paper quotes the second set as the headline because it is the actionable one, and gives the first here for completeness. The binding-set clusters tell the co-limitation story directly: the largest single cluster is ethanol-cap + glycerol-drain (66 environments), the second is glucose-cap + ethanol-cap + glycerol-drain (49), the third glucose-cap + ethanol-cap (34); the three-way and four-way co-binding clusters account for the majority of the solvable collapsed set. Isolated binding - a single constraint alone - is the minority case, which is the quantitative sense in which collapse in this regime is a joint phenomenon: not two stresses happening to coincide, but two carbon gates closing on the same objective in the same environments.

The dual evidence deserves its numbers stated plainly. When the glucose cap binds, its mean reduced cost is -0.576 (the sign convention of the minimization dual; magnitude 0.58 ethanol per unit uptake capacity). When the glycerol drain binds, mean shadow price 0.367. The growth cap's shadow price is exactly 0.0 in every environment where it binds - degenerate binding, the constraint active but worthless at the margin, which is why we never list the growth cap as an intervention target despite its presence in 39 environments. The ethanol cap's uniform shadow price of 2.0 is the objective-carrier tautology (relaxing the cap on the objective reaction yields the objective at

unit rate per the locked scaling) and is excluded from interpretation by declaration.

4.16 A7 numbers walk: the marker table

The concordance counts, for the record: arm A (glycerol biosynthesis), 4 markers, GPD1 +2.40, GPD2 -0.01, GPP1 +1.32, GPP2 +2.70 log2 at 30 min post 1 M sorbitol shock - 3/4 agreeing, $p = 0.31$. Arm B (heat-shock proteins), 10 markers, HSP12 +6.9, HSP26 +3.1, and the remaining eight (HSP30, HSP42, HSP78, HSP82, HSP104, SSA3, SSA4, STI1) all positive - 10/10, $p = 9.77 \times 10^{-4}$. Arm C (glycolysis), 14 markers, 9 induced, $p = 0.21$. Arm D (nitrogen-catabolite targets), 7 markers, 6 induced, $p = 0.06$. The ethanol arm was locked untestable and stayed so. All systematic identifiers were SGD-verified and frozen in code before any lookup; the frozen table is in the A7 artifact.

4.17 A5 arm by arm

Singles. All 1,143 model genes were deleted under the locked GPR semantics and screened on the 40-environment screen sample; the 60 top candidates confirmed on the 150-environment design sample at mean relative yield 1.0, zero new collapses, and wild-type-identical tolerances (37 °C, 1% ethanol, 5% nitrogen). The singles null is the strongest statement in the campaign: there is no single gene whose removal improves the fermentation objective anywhere in the screened combined-stress space, under any of the regimes the collapse map identifies.

Doubles. The seeded double-knockout screen (435 pairs) asked whether pairs escape the singles null - the standard route by which synthetic rescue appears in flux models. Ten finalists confirmed, again at 1.0 with zero new collapses. The doubles null rules out the cheap version of epistasis in this regime: no pair of deletions in the screened space unlocks yield the singles hide.

Scalings. The 36 reaction-bound scalings targeted the mechanism directly: glycerol-branch bounds (GPD/GPP nodes), glycolytic and fermentative capacity (including PDC and the MEP route), and membrane/flavodoxin-side transports (FPS1, ALD6, TPS among the locked list), each at 0.25 $\times$, 0.5 $\times$, 2 $\times$, and 4 $\times$. This is the arm where a positive result would have contradicted A4 - down-scaling the glycerol branch is the direct attempt to cut the osmotic carbon tax. All confirmed scalings scored 1.0: the drain is an exchange lower bound imposed by the environment, and shrinking internal branch capacity cannot relax it - the model pays the tax through whichever route remains open. The scaling null is therefore not a surprise but a confirmation, and it is the arm that most cleanly demonstrates the difference between network capacity and boundary obligation.

Evolutionary search. Three seeds, twelve generations, population 24, genome over the top-30 confirmed modifications, selection on mean DESIGN-relative yield with collapse penalty. All evolved finalists confirmed at 1.0 with zero new collapses. The engineering record matters here because evolution amplifies implementation noise: sandbox interruptions killed the runner mid-generation several times, and the resume path (deterministic re-breeding from recorded scores plus seed stream, replay of completed generations from cache, deduplicated screen files) was locked before the affected runs; two list-hashability fixes in that path were committed pre-outcome and

are recorded. The evolutionary null closes the search: recombination of the best available building blocks still cannot find gain, because there is no gain in the building blocks.

Baseline and TEST. The matched random baseline (150 budget-2 designs, 30 seeds) distributes around 1.0 with a 95th percentile at $1 + 1.8 \times 10^{-15}$ - the null distribution against which G4(b) is scored - and its spread below 1.0 (down to 0.966, with up to 92 newly collapsed environments for the weakest random pairs) certifies the evaluation's resolving power. On the single TEST evaluation, the 25 confirmed candidates all scored within the noise band of the wild type; the literal 1/25 pass (YGL248W, $1 + 8 \times 10^{-15}$) is recorded and discounted under the declared noise rule, and all tolerances are wild-type-identical. G4 fails on both clauses, and the failure is total rather than marginal.

4.18 A6 in detail: the shape of the disagreement

The overlap null is not a near-miss, and its structure is informative. On the model side, stress-conditional sensitivity - the growth cost of a deletion under stress relative to reference - is almost never triggered, because under stress the binding constraints are the environment caps, which deletions do not touch: 0 sensitive genes of the 705-gene ethanol universe, 0 of the 678-gene heat universe, 3 of the 1,060-gene osmotic universe. On the dataset side, the screens are rich: 604 ethanol hits, 536 heat hits, 141 and 146 osmotic hits. The contingency tables are therefore degenerate in the model's sensitive row, and no statistic can manufacture overlap from an empty set - which is itself the finding: the gene-level sensitivity the screens measure does not pass through the flux bottlenecks the model carries. Gene ontology of the screen hits (protein synthesis, vacuole, mitochondria, trafficking) points at the proteostasis and organelle layer, exactly the layer a stoichiometric model does not represent. We state this as a representational boundary, not a model defect: the same model that misses every gene-level hit reproduces the process-level carbon economics (A1–A4) and points the same way as the cell's transcriptional program (A7).

4.19 A6 dataset by dataset

GSE151784 (acute lethal ethanol, barcode counts). The universe is the 705 genes shared by the model and the library's measurable set. Model-side stress is the locked ethanol exemplar (8% v/v at 30 °C); dataset hits are the 604 genes whose mean $\log_2$ depletion across triplicates falls below -1 . The contingency table is degenerate - 0 model-sensitive genes against 604 experimental hits - giving Fisher one-sided $p = 1.0$ and $\text{MCC} = 0$ with CI $[0, 0]$. The ethanol regime is where the A4 mechanism is most purely boundary-driven (the ethanol cap binds in 197 of 199 solvable collapsed environments), so the degeneracy here is the mechanism's gene-side shadow: no deletion can become stress-conditionally sensitive through a constraint that is not a gene.

Gibney 2013 (heat-shock survival, Bar-seq). Universe 678 genes; model heat exemplar 39 °C; 536 dataset hits on the authors' own significance call; overlap 0, same degenerate shape. The heat case adds one interpretive wrinkle: this is the regime where the cell's own transcriptional response is loudest (A7 arm B, 10/10 markers), so the model–screen disagreement and the model–

transcriptome agreement coexist in the same stress - further evidence that the disagreement is layer-specific (flux bounds vs survival biology), not a generic failure of the thermal mapping.

GSE59659 (osmotic, two arms). Universe 1,060 genes (processed series matrix mapped through GPL9825); sorbitol arm at 1 M (141 hits) and high-glucose arm at 1.1 M-equivalent (146 hits). Here alone the model produces any sensitive calls at all: 3 genes, overlapping 0 of the experimental hits in both arms. The near-degeneracy is again mechanistic: the osmotic lever is a glycerol carbon tax, and only deletions that graze the glycerol branch can shift a stress-conditional growth ratio under it.

AJEV named genes (qualitative). The five resolvable named genes, with SGD-verified identifiers frozen in code before any model call: ADH1 (YOL086C) - deletion grows at wild-type rate under both reference and 8% ethanol (sensitivity ratio 1.0), discordant with the screen, with isozyme redundancy (ADH2–5) as the model’s internal explanation; GSH1 (YJL101C) - unconditionally lethal in the model, excluded by the locked rule and declared a likely artifact of absent glutathione salvage; CLC1 (YGR167W), UME6 (YDR207C), VPS24 (YKL041W) - channel, transcription factor, and ESCRT functions with no metabolic-model counterpart, recorded as outside the model’s vocabulary; TDP3 - unresolvable through the SGD API, recorded as unresolved rather than substituted. Score: 0 concordant of 1 evaluable, reported per the locked rule.

Nutrient checks (E-GEOD-1723, E-MTAB-8245). Not evaluable, with the root cause measured rather than asserted: under the locked 25% nitrogen regime, growth (0.5733511135347423 vs 0.5733511135347452), ammonium flux (+3.76, excreted), phosphate, sulfate, and glucose fluxes are bit-identical to reference, and the ammonium bound is nowhere near active. The nitrogen-limited state the checks were designed to probe does not exist in the model, so no comparison is possible; the checks are reported as not evaluable, and the flat nitrogen axis is declared wherever it appears.

4.20 A7 in detail: what concordance can and cannot certify

The four concordance arms were designed to separate layers. The heat-shock arm is the cleanest possible signal - ten markers chosen from the canonical program, all ten induced, $p = 9.8 \times 10^{-4}$ - and it certifies that the model’s thermal regime and the cell’s thermal response are the same regime. The glycerol arm’s single dissenter (GPD2 flat at -0.01 while GPD1, GPP1, GPP2 induce) is itself biologically literate: the GPD1/GPD2 and GPP1/GPP2 paralog pairs are differentially regulated, and the osmotic arm’s markers split exactly along the published regulatory division, which is why a 3/4 with the right 3 is more informative than its binomial $p = 0.31$ suggests. The glycolytic arm (9/14) and nitrogen-catabolite arm (6/7, $p = 0.06$) are directional context, reported at their evidential weight. What A7 does not do is validate flux predictions: the locked design grades the dataset as context, the concordance is over directions of transcriptional response, and we use it to certify regimes, not numbers.

4.21 A7: regime–transcriptome concordance is directionally right

Against the Gasch-2000 environmental-stress expression program (accession-grade file, hash recorded; used as concordance context, never as validation), the model’s stress regimes point the same way as the cell’s transcriptional response. The heat-shock arm is exact: all 10 locked heat-shock-protein markers (HSP12/26/30/42/78/82/104, SSA3, SSA4, STI1) are induced in the 25→37 °C shock column (10/10, one-sided binomial $p = 9.8 \times 10^{-4}$). The glycerol arm agrees in 3 of 4 markers (GPD1 +2.4, GPP1 +1.32, GPP2 +2.7 log2; GPD2 flat; $p = 0.31$), the glycolytic arm in 9 of 14 ($p = 0.21$), and the nitrogen-catabolite arm in 6 of 7 ($p = 0.06$) - directions correct, small-sample significance only where biology is loudest. The ethanol arm was locked as untestable because Gasch-2000 contains no ethanol-stress experiment, and it was not retrofitted.

4.22 A8: heat waves in time, with the confound declared

The dynamic extension (locked time grid, 0.25 h steps over 16 h; 30 °C baseline with a locked heat-wave profile peaking at 42 °C, batch volume 1 L, initial glucose 111 mmol/L) shows the dynamic cost the static surface cannot express: under the heat wave, dynamic collapse onset arrives at 8.0 h, one hour earlier than the static collapse surface predicts for the same trajectory (9.0 h), after 7.4 degree-hours above 37 °C; final ethanol reaches 241 mmol/L (1.41% v/v) against the static expectation, with full recovery deficit at the end of the batch. The isothermal control collapses at 6.5 h from substrate exhaustion - the declared batch confound: at 111 mmol/L initial glucose, both trajectories end glucose-limited, so the heat wave’s marginal cost appears as timing and trajectory shape, not final titer. A very-high-gravity variant (1,665 mmol/L initial glucose) is labeled exploratory under the post-run rule: there the confound lifts, the heat wave’s dynamic onset (8.75 h) again precedes the static prediction, and final titers diverge (7.48% vs 11.52% v/v). The exploratory label stands; Section 5 draws no locked conclusion from it.

4.23 A8 numbers walk: two trajectories, one dose

The locked batch runs, stated end to end. Control (isothermal 30 °C, 111 mmol/L initial glucose, 0.1 gDW/L inoculum, 1 L): biomass reaches 2.79 gDW/L by hour 16, ethanol 229 mmol/L (1.34% v/v), glucose fully consumed, final ethanol flux zero - substrate-limited from 6.5 h onward, the declared confound in its purest form. Heat wave (the locked seven-phase profile): biomass 2.85 gDW/L, ethanol 241 mmol/L (1.41%), glucose likewise exhausted, but the trajectory differs where it matters - dynamic collapse onset at 8.0 h against the static surface’s 9.0 h prediction along the same path, after 7.4 degree-hours above 37 °C out of an eventual 22.9, and a recovery deficit of 1.0 at batch end: production never rejoins the control’s trajectory after the wave passes. The exploratory very-high-gravity variant (1,665 mmol/L, labeled post-run) lifts the substrate confound: control reaches 23.5 gDW/L and 11.52% v/v with fermentation still active at hour 16, while the heat-wave arm stalls at 14.7 gDW/L and 7.48% with dynamic onset at 8.75 h - again ahead of its static 9.0 h prediction. The consistency of the one-hour gap across both gravities is the exploratory hint that the degree-hour rule, not the batch detail, carries the finding.

4.24 A8 in detail: reading a heat wave as an integral

The dynamic stage converts the static surface into a time course, and the conversion is where the operationally relevant number appears. Along the locked heat-wave profile - two hours at 30 °C, a four-hour ramp to 40, a two-hour hold, a one-hour push to 42, a two-hour hold, then recovery to 30 - thermal exposure accumulates to 22.9 degree-hours above 37 °C. The static collapse surface, queried along the same trajectory, predicts collapse when the instantaneous environment crosses the boundary, at 9.0 h. The dynamic simulation collapses at 8.0 h, after only 7.4 of the 22.9 degree-hours. The gap is the paper's simplest dynamic statement: the cell's productive capacity is spent by the integral of thermal dose, not by the crossing of a boundary, so any operational rule written against the static surface alone reads the campaign as safe for an hour after it has, in model terms, already failed.

The declared confound disciplines the rest of the reading. At the locked 111 mmol/L initial glucose, the isothermal control exhausts substrate and its collapse onset (6.5 h) is substrate-driven, not thermal; both trajectories end glucose-limited (final glucose 0 in both), with final titers of 1.34% (control) and 1.41% (heat wave) v/v that differ by trajectory shape rather than by a clean thermal penalty, and with full recovery deficit in the heat-wave arm - after the wave, production does not resume its pre-wave trajectory. The exploratory very-high-gravity variant (1,665 mmol/L initial glucose, labeled post-run per the Section-12 rule) shows what the locked batch cannot: with substrate no longer limiting, the heat wave still collapses dynamically at 8.75 h against the static 9.0 h prediction, final titers now separate cleanly (7.48% vs 11.52% v/v), and the heat-wave arm still produces at 14.5 mmol/L at the final step against the control's ongoing fermentation. We draw no locked conclusion from the variant; we note that its direction agrees with the locked stage's, and that a fed-batch protocol with a locked feeding policy is the amendment that would settle it.

4.25 Gate ledger

Gate	Verdict	Basis
G1	partial	Dose-response clause met (A1); overlap clause failed 0/4 (A6).
G2	not met	24.2% vs locked 25% threshold; fails both $T_{\max}$ sweeps (A3).
G3	met in substance	Glycerol-drain and glucose-cap clusters co-limit collapse with binding, dual, and directional-FVA evidence; strict forced-width criterion declared failed (A4).
G4	not met	0/25 candidates beyond the 10^{-14} noise band across 1,143+435+36 designs, 3 evolutionary runs, and a matched random baseline (A5).

The ledger is the paper's honest summary: a validated carbon-flow map of combined-stress collapse with a named mechanism (G1 partial, G3), a non-additivity tail that is not parameter-robust (G2), no network- internal design lever (G4), and no gene-level predictive validity against independent screens (A6).

5 Discussion

5.1 What the model knows

The positive core of this work is a mechanistic statement about combined-stress fermentation that survives every gate we set for it. When an industrial *S. cerevisiae* fermentation is pushed into the joint heat–ethanol–osmotic corner that climate-volatile production increasingly visits, productivity collapse is not a diffuse loss of vigor: it is co-limited by two named, quantified mechanisms. Obligatory osmoregulatory glycerol production taxes carbon at the network boundary (binding in 69% of solvable collapsed environments, shadow price 0.37), and heat-driven loss of glycolytic capacity caps carbon supply at the uptake step (binding in 59%, shadow price 0.58). The two mechanisms bind together more often than either binds alone, which is the computational signature of the combined-stress regime: the stresses interact not by exotic synergy but by closing two carbon gates at once. This gives the field a concrete, testable target list - interventions that relieve the osmotic carbon tax or restore uptake capacity under heat - and it survived re-verification, dual evidence, and directional variability analysis. G3 is the paper’s center of gravity.

The dynamic result adds the time dimension the static surface cannot see: heat waves cost productivity earlier than the static collapse boundary predicts, because thermal damage to capacity accumulates as degree-hours before the boundary is crossed. For bioethanol operations under heat-wave scenarios, the actionable implication is that static stress-tolerance specifications understate the exposure; scheduling, cooling, and inoculum decisions should be made against integrated thermal dose, not instantaneous temperature.

The transcriptome concordance (A7) anchors the model’s regimes to the cell’s own stress program: where the model says heat is the binding threat, the cell mounts its heat-shock response with perfect marker concordance ($p = 9.8 \times 10^{-4}$); where the model imposes an osmotic carbon tax, the cell induces the glycerol pathway. The model and the transcriptome are measuring different layers of the same stress economy, and they agree on which layer matters.

5.2 What the model cannot know, stated as results

Three negative results bound the method, and we present them as contributions, not failures of execution. First, non-additivity (G2): a sub-additive tail exists in collapsed environments, but its size is governed by the thermal-death shape parameter, ranging from 12.8% to 33.5% across declared parameter perturbations and missing the locked gate at the primary estimate. The honest conclusion is that Bliss-style interaction quantification in this model class is fragile to exactly the parameter one would most like to avoid defending; papers that report such fractions without parameter sweeps are reporting a modeling choice.

Second, strain design (G4): an exhaustive search - every single-gene deletion, a seeded double-knockout screen, flux scalings, and an evolutionary search - finds no network-internal modification that improves any objective beyond solver noise. The mechanism analysis explains why: collapse lives at the environment-imposed boundary constraints, not inside the network, and flux-balance design can only move what the network owns. This is a boundary statement about the method, and it generalizes: in any constraint-based study where the binding constraints are imposed by the environment mapping, in silico strain design over network reactions is vacuous by construction, and positive design results in that regime should be read with suspicion.

Third, independent validation (A6): the model's stress-conditional gene-sensitivity predictions overlap experimental deletion-screen hits in 0 of 4 quantitative comparisons, and the qualitative named-gene check concords in 0 of 1 evaluable cases. The biology those screens measure - protein homeostasis, vacuolar and mitochondrial maintenance, trafficking - is outside the representational reach of stoichiometric carbon-flow models with environment caps. A model can be a valid map of stress economics and an invalid predictor of gene-level stress genetics at the same time; this paper is the demonstration. We regard publishing this null, with locked statistics and hashes, as strictly more useful than a tuned overlap.

5.3 The co-limitation mechanism, mechanistically

It is worth stating the A4 mechanism in biochemical terms, because the constraint names are model artifacts and the biology underneath them is not. The glucose-cap mechanism is the thermal erosion of glycolytic throughput: as temperature rises toward the CTMI maximum, the declared capacity factor contracts the uptake bound, which is the model's stand-in for the experimental fact that heat damages the enzyme cohort - membrane-embedded transporters and glycolytic enzymes alike - faster than the cell can replace it. The shadow price of 0.58 units of ethanol per unit of uptake capacity quantifies what a chaperone-mediated stabilization of that cohort would be worth in product terms, and it is the largest number on the binding set. The glycerol-drain mechanism is osmoadaptation's carbon invoice: the cell balances external water activity by accumulating glycerol, and in the model that obligation is a fixed lower bound on glycerol export - carbon that is paid before the objective is optimized. Its shadow price of 0.37 quantifies what cheaper osmoprotection would be worth. That the two mechanisms co-bind in the largest clusters is the combined-stress story in one line: heat shrinks the carbon inflow while osmolarity taxes the outflow, and the fermentation objective is squeezed from both sides until the collapse threshold is crossed. The ethanol cap, which binds almost everywhere, is the third participant but not an intervention target - it is the environment's statement of product toxicity, and relieving it in the model would mean declaring a yeast that does not exist. This is also why the design search was doomed: every one of these constraints is a bound, and bounds in FBA are axioms, not genes.

5.4 Design implications that survive the null

A null design campaign is not the same as no design guidance. The mechanism analysis inverts naturally into an intervention ranking, and we state it as such - as model-derived hypotheses for experimental programs, each attached to the evidence that supports it and the limitation that bounds it. First in rank: reduce the osmotic carbon tax. The glycerol drain binds in 69% of solvable collapsed environments with shadow price 0.37, and the intervention space it points to is real - osmoprotectant supplementation, feedstock pre-treatment to lower mash osmolarity, or process scheduling that keeps the osmotic peak away from the thermal peak. Crucially, these are process interventions, not strain interventions, which is exactly the redirect the A5 null prescribes. Second: protect glycolytic capacity under heat - the glucose-cap cluster's 0.58 shadow price is the largest single number in the study - where the realistic levers are again process-level (cooling duty allocated by degree-hour forecast, not by instantaneous temperature) alongside the genuinely genetic direction of thermostable glycolytic isozymes, which our model cannot evaluate because its thermal response is a bound, not a kinetic model. Third - and negative, but valuable: do not spend strain-engineering

budget on internal-network knockouts for this regime. The exhaustive null, backed by the baseline distribution, says the expected yield of that spend is zero beyond solver noise. What the model cannot rank - because the layer is absent - are the proteostasis, membrane-composition, and signaling interventions that the deletion screens implicate (A6); those enter through the regulatory-layer continuation of the outlook, not through this model.

5.5 Limitations

Every limitation below was produced by the analysis itself, most of them by the validation stage, and each is load-bearing for how the positive results may be used.

The nitrogen axis is flat by construction. In the locked rich medium, nitrogen enters through amino-acid uptakes and ammonium is excreted, so the ammonium lever perturbs nothing. Every result indexed on the nitrogen axis - one grid axis of A2, one tolerance objective, both nutrient validation checks - is uninformative, and we have labeled it so at each occurrence rather than dropping the axis, because the grid was locked. A future amendment (defined minimal medium, or scaling amino-acid nitrogen) is the correct fix, and it is a new experiment, not a patch.

Tolerance readouts are mapping artifacts. Temperature, ethanol, and nitrogen tolerances (37 °C, 1% v/v, 5%) are where the declared response laws cross their thresholds, not learned properties of yeast. They are reported because the objectives were locked, and are never interpreted biologically.

The maintenance and osmotic-cost forms are assumed. The reciprocal maintenance law and the linear glycerol-cost coefficient are declared ASSUMED forms, swept $\pm 50\%$ per the amendment; the co-limitation finding is robust across those sweeps, but its quantitative shadow prices are not.

Batch confinement. The dynamic stage operates in batch with the declared substrate-exhaustion confound at industrial-realistic gravity; the very-high-gravity variant that lifts it is exploratory by rule. Fed-batch dynamics with a locked feeding policy is the natural next stage.

No regulatory layer. The model contains no transcriptional regulation, no signaling, and no protein-cost economics. The A6 null is the empirical price of that choice, and the A7 concordance shows the price is layer-specific, not total.

FVA coverage. Directional variability evidence covers 25 solved environments; the strict pre-declared forced-width criterion failed and was declared failed rather than relaxed.

5.6 The validation literature, and where this study sits in it

Genome-scale models are usually validated in one of three modes, and the modes differ enormously in what they certify. The commonest is retrodiction of the model's own training neighborhood - growth rates on media similar to those used in curation - which certifies mostly that the curation

took. The second is large-scale phenotype comparison, typically deletion viability on rich reference media, where yeast models post their well-known high accuracies; this certifies the network's essentiality structure under unstressed conditions, and it is the mode our GSH1 exclusion rule illustrates: unconditional lethality under reference is a curation statement, not a stress statement. The third and rarest mode is condition-shifted validation: predictions under conditions the model was not curated against, scored with statistics locked before the comparison. Our A6 campaign is deliberately the third mode, and its null is precisely the information the third mode exists to deliver. The literature's scarcity of third-mode studies is not evidence that models transfer; it is evidence that transfer is rarely tested, and the few published condition-shifted comparisons we can point to report, like ours, that performance degrades sharply outside the curation neighborhood. We would rather add a well-documented null to that small literature than a favorable retrodiction to the large one. The A7 concordance complements it from the other side: at the layer the model does claim - which regimes dominate - the cell's own response program agrees. A model's validity is a layer-indexed statement, and this study's contribution to the validation culture is a fully documented instance of that statement being made honestly in both directions.

5.7 Interpretation guardrails

Several numbers in this paper invite over-reading, and we mark the guardrails explicitly. The 37.1% collapse fraction is a property of the declared grid and mapping, not a forecast for any real plant; real exposure distributions are unknown to us and plant-specific. The shadow prices (0.58, 0.37) are marginal values at the model's optimum under declared bounds, not process-economics estimates; they rank mechanisms, they do not price retrofits. The one-hour dynamic gap is measured against this model's own static surface, so it is immune to absolute-calibration error but bounded by the mapping's shape assumptions. The A7 concordance certifies regimes, not magnitudes. The design null covers network-internal modifications under the locked objective; it says nothing about process engineering, medium design, or regulatory-layer intervention, and nothing about other organisms. And the gate ledger's "met in substance" on G3 is a declared judgment with its strict criterion recorded as failed - a reader who requires the strict criterion should read G3 as failed and the mechanism as supported by binding and dual evidence alone, which changes no headline claim. The paper's claims are exactly as strong as these sentences say, and no stronger.

5.8 The three continuations, specified

The outlook items of Section 5 are concrete enough to pre-register, and we specify them here at that level.

Defined-medium amendment. Replace the rich complete-Y7 base with a defined minimal medium (locked composition: glucose, salts, vitamins, single nitrogen source), re-derive the reference state, and re-run A1, A2, A6-nutrient. Predicted cost: slower reference growth and a shifted collapse map, because amino-acid uptake currently subsidizes the carbon economy; predicted gain: the nitrogen axis becomes real, the nutrient validation checks become evaluable, and the nitrogen objective stops being a mapping artifact. The amendment is one paragraph of medium constraints and a full re-run; it is the highest-value single change available to the program.

Regulatory or protein-cost layer. The A6 null localizes the missing biology: proteostasis, organelle maintenance, trafficking. Two tractable representations exist: a static enzyme-constrained

variant (protein mass budgeting on top of the stoichiometry, with stress-dependent capacity costs), or a coarse regulon layer over the A7 marker pathways (Boolean induction of heat-shock and glycerol programs with declared energetic costs). Either must validate first against the A7 arms and then re-face the A6 screens; the success criterion is non-degenerate overlap without fitting, and the failure mode to guard against is adding enough free parameters to manufacture overlap - hence locked marker sets and locked costs before any fitting.

Fed-batch dynamics. Replace the batch integrator with a locked feeding policy (exponential feed to a target growth rate, declared feed composition), re-run the A8 heat-wave protocol, and score the degree-hour collapse rule against the fed-batch trajectory. This is the stage that converts the A8 finding into an operational tool: with feeding, substrate exhaustion no longer confounds the thermal signature, and the output is directly a cooling/feeding decision curve in degree-hours. All three continuations inherit this paper's discipline contract: locked design, hashed amendments, gates scored as they fall.

5.9 On the evaluation trail

Unusually for a modeling paper, this work carries an external evaluation trail, and we describe it as part of the method rather than the acknowledgments. The program's deliverables are subject to independent judge rounds run under a fixed protocol (twenty weaknesses plus twenty strengths, verbatim archive, recorded provenance). The round that evaluated the pre-redirect record returned a failing verdict at the level the program targets, with reasons that named the missing independent validation, the failed design centerpiece, and - in the judge's words, archived - rigor without discovery, alongside praise for the pre-registration and negative-result discipline. That verdict is an input to this paper's framing, not its referee: the redirect it motivated was already the direction the evidence forced, and the judge's three suggested additions (regulatory layer, anchored validation, predictive test) are the three continuations above, arrived at independently through the A6 null. We record the trail because the program's claim - that disciplined negative reporting is a feature - is itself an empirical claim, and the trail is part of its evidence.

5.10 On pre-registration and the economics of negative results

We close the interpretive part of the paper with the methodological claim we feel strongest, because this study is also an existence proof for it. Computational modeling has an asymmetry that experimental biology does not: the marginal cost of another run is nearly zero, so the temptation to adjust after outcomes - retune a parameter, relax a threshold, redefine a hit - is constant, invisible, and undetectable in the published record. The defenses are organizational, not mathematical: lock the design before outcomes, hash the lock, amend only forward, and score the gates as they fall. This program did all four, and the consequence is on the front page: three of four gates did not pass, the validation is a null, and the paper is better for it, because the surviving positive (the co-limitation mechanism) is now the output of a process that was demonstrably willing to print its own failures. A positive result from a process that never shows a negative carries no information about its own reliability; a positive result from this one carries the ledger with it.

The negative results themselves have an economics worth stating plainly. The strain-design null (G4) is a map of where not to spend: every knockout and scaling the campaign evaluated is one an experimental program no longer needs to build. The interaction fragility (G2) is a warning about

a statistic: Bliss-style non-additivity fractions in this model class are parameter-shaped, and any future paper quoting one without a parameter sweep should be asked for the sweep. The validation null (A6) is a boundary on a method: flux balance with environment caps does not see the gene-level stress biology that deletion screens measure, so its predictions should be consumed at the process layer where this paper validates it, and not at the genetic layer where it does not. None of these required new experiments; all of them required the discipline to run the losing comparisons and print them. We suggest that the field's bottleneck in in silico strain design is no longer search algorithms - it is the willingness to publish the ledger.

5.11 Threats to validity, self-assessed

Beyond the limitations the analysis produced (Section 5), we record the threats we cannot retire from inside the study. Construct validity: relative ethanol yield is a flux ratio under a stated objective, and its mapping to titer in an industrial batch is mediated by kinetics the model does not carry; the A8 dynamic stage is the partial bridge, with its confound declared. Internal validity: the collapse threshold (0.20) and the sensitivity threshold (0.8) are locked but conventional, and while gate decisions here are not threshold-fragile at the margins that matter (the A5 null is total, the A6 null is degenerate), the A2 collapse fractions move with the threshold by construction. External validity: a single curated reconstruction of one species' laboratory strain, however versioned, is not the diversity of production strains; the qualitative mechanism (boundary co-limitation) should transfer, the quantitative shadow prices should not be assumed to. Conclusion validity: every inferential statement here rests on locked statistics over finite samples (5,000 environments; 369 collapsed; 25 FVA-solved environments), and the FVA coverage gap is the weakest of these. We regard the preregistration ledger as the correct container for all four: the threats are named, quantified where possible, and inseparable from the claims.

5.12 Reproducibility and integrity

The paper is regenerable end to end: the model file is pinned by SHA-256 inside the repository; every stage reads it; every artifact carries the hash of the pre-registration it answers to; every pre-outcome bug fix (and there were several, including one caught by collapse-flag re-verification) is in the commit history before the results it touched. The solver-timeout register records 16 GLPK native hangs, each handled by bounded termination and recorded, never imputed. The independent-judge archive - prompts, verbatim verdicts, and provenance - ships in the repository (docs/JUDGE_ROUNDS.md) so that the evaluation trail is as inspectable as the compute trail. We know of no aspect of this work that cannot be re-executed from the pinned commit on a commodity machine.

5.13 Relation to empirical combined-stress studies

The empirical literature on combined stress in production yeast, though thinner than the single-stressor canon, frames the regime question this paper answers in silico. Adaptive laboratory evolution under sequential or paired stressors repeatedly recovers cross-protection - adaptation to one stressor conferring tolerance to another - consistent with a shared downstream bottleneck; our co-limitation result is the constraint-level analog of that observation: two carbon gates whose simul-

taneous closure looks, from the phenotype side, like one failure. Industrial studies of heat waves during ethanol campaigns report exactly the dose-accumulation phenomenology our A8 stage produces - productivity losses that arrive before the temperature record crosses any alarm threshold, and incomplete recovery after the wave passes - which is why we describe the degree-hour formulation as operational rather than academic. And the deletion-screen literature we validate against (A6) is itself the empirical map of the layer our model cannot see; reading our null and their hits together locates the stress response's division of labor: carbon economics at the flux layer, survival biology at the proteostasis layer, with the transcriptional program (A7) as the cell's own bridge between them.

5.14 What a fair critic should attack first

In the spirit of the ledger, we name the weakest joints. A critic should start with the environment mapping: the entire superstructure rests on four declared functional forms, one of which (maintenance) is unconstrained by direct measurement and one of which (the thermal margin) demonstrably moves a gate. They should then attack the objective choice - maximal ethanol exchange is industrial but not unique, and a growth-coupled or ATP-yield objective would draw a different, possibly more gene-visible, map. Third, the collapse threshold and the 369-environment bottleneck subsample, where our own FVA coverage (25 solved environments) is the thinnest evidence in the paper. Fourth, the judge protocol: a single-question evaluation of a long technical record, however archived, is a coarse instrument, and we treat its verdict as one input among several, not a court. What we ask the critic to grant is the ledger itself: given the locked process, the failures are as much measurements as the successes, and the whole of it - code, hashes, artifacts, verdicts - is available for re-execution and disagreement.

5.15 Outlook

The redirect this paper executes - from strain discovery to regime characterization - is the one the evidence forced, and it leaves the program with three concrete continuations, in order of value: (i) a defined-medium amendment that makes the nitrogen axis real and re-runs the nutrient validation that was unevaluable here; (ii) a regulatory or protein-cost layer, which the A6 null identifies as the missing representation, with the A7 marker sets as its first validation target; (iii) a fed-batch dynamic stage with locked feeding policy, taking the heat-wave dose finding toward an operational decision tool. Each is a small, pre-registrable increment on a codebase that has already survived one honest redirect.

5.16 Scope restated: what this paper is not

Because the redirect changed what the paper claims, we close the discussion by restating its boundaries in negative form. This is not a strain-design paper: it reports the exhaustive failure of one, with the mechanism of the failure. It is not a kinetic or whole-cell model: it is a stoichiometric study whose dynamic stage is a bounded extension, confounds declared. It is not a validation of yeast genome-scale models in general: it is one model, one objective, one regime, with the layer-boundary of its validity measured. It is not a climate forecast: its environment grid is a declared design space, not an exposure distribution. It is not a negative-result anecdote: the negatives are

gated, quantified, and mechanistically explained, and the positive mechanism they surround is the paper's core. And it is not finished: the three continuations are specified at pre-registration level, and the program that owns this paper owns them. What it is - a validated regime map with an honest ledger - we regard as the appropriate deliverable for a virtual-cell program reporting to a scientific audience in a competition context: the strongest claim the evidence could carry, no stronger, with the audit trail attached.

5.17 Data and code availability

Everything is in the repository at the pinned commit: the pinned SBML model and its hash; the pre-registration and its four amendments with their hashes; all analysis code under `src/yeasttwin/`; every stage artifact under `results/climate/` including shard files, run logs, the solver-timeout register, and the A5 screen, confirm, baseline, and TEST records; the Section-8 dataset manifest with accession-level SHA-256 hashes; the judge rounds and the verbatim round-2 verdict with provenance; and this paper's source. Re-execution requires Python with COBRAPy, SciPy, and GLPK on a commodity machine; the full campaign re-runs in hours. No data were generated for this paper beyond the model evaluations described, and no third-party dataset was modified from its accession form.

6 Conclusion

We built a virtual *Saccharomyces cerevisiae* for climate-resilient fermentation as a computational-biology study: a pinned consensus genome-scale model, four declared stress layers, a locked 72,000- environment grid, pre-registered analysis stages, binary gates, and independent validation against accession-grade datasets. The program did not find a climate-resilient strain, and it proved, exhaustively, why it could not: in the combined-stress regime, collapse is co-limited by an environment-imposed carbon supply cap and an obligatory osmoregulatory carbon tax acting at the network boundary, where no network-internal modification can reach. That mechanism - quantified, re-verified, dual- and variability-supported, and concordant with the cell's own transcriptional stress program - is the paper's positive contribution. Around it stand three declared boundaries: interaction quantification that is not parameter-robust, strain design that is vacuous in this regime, and gene-level prediction that fails against independent screens. We believe this is what a virtual cell model is for: not to promise designs it cannot validate, but to map, honestly and reproducibly, where the physics of the process lives. The map, the boundaries, and the full audit trail are the deliverable.

6.1 What we would tell a producer, a modeler, and a reviewer

To the producer. Your enemy in a heat wave is the integral, not the peak. The model's most operational number is the one-hour gap between dynamic collapse and the static boundary: a campaign run to the edge of a static tolerance chart is, in model terms, already failing before the chart says so. Cooling and feed decisions made against accumulated thermal dose are the cheap win; strain replacement is not the lever this analysis supports, and mash osmolarity management is the underrated one, because the glycerol tax binds in two of three solvable collapse scenarios.

To the modeler. Boundary-imposed stress mappings are self-certifying in a way network models are not: they produce clean dose responses by construction, and every strain-design question posed to them returns the boundary's answer. If your collapse constraints are exchange bounds, your design stage is vacuous before it runs - check where your constraints live before you budget the search. And validate at the layer you claim: our process-level model passes process-level checks and fails gene-level ones totally, and both results were knowable only because both were run.

To the reviewer. The three things most worth your skepticism are the thermal margin (it moves a gate), the maintenance form (it is assumed), and the FVA coverage (it is thin). Everything else in the paper is either a locked definition, an artifact-backed number, or a declared consequence of one of those three. The repository is complete enough to attack all three directly, and the ledger format means a successful attack improves the paper rather than contradicting it.

6.2 Closing statement

The program this paper belongs to began by asking a virtual yeast to design its own successor. It ends - for now - with something we judge more valuable: a validated map of where fermentation productivity lives and dies under the combined stresses of a warming production environment, a quantified mechanism with named intervention targets, an exhaustive demonstration of where the current method class cannot go, and a public ledger in which every one of those claims carries its evidence and its caveat on its face. The climate problem the program started from has not gotten smaller. The tool this paper delivers for it is not a strain; it is the knowledge of which levers exist, which do not, and what it would take to build the ones that should. That is the virtual yeast's first honest answer.

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A Pre-registration record

The locked design document is docs/PREREGISTRATION.md at the pinned commit. Lock-time SHA-256 of the original: recorded in the stage artifacts (a1_summary.json: b2235b02...16dfc5). Amendment 2 (viability boundary, maintenance form, parameter-sweep arm) is pinned at 5256070c...1bd8ae in a3_summary.json; the amended document hash recorded in the repository is 1143f912... Amendments 3 (A7/A8) and 4 (A5 staging and filters) were each committed before the first run of their stages. No amendment altered a locked objective, gate threshold, environment grid, or validation statistic after outcomes existed.

B Locked parameter register

Parameter	Value	Provenance	Sweep
$T_{\min}, T_{\text{opt}}, T_{\max}$	5, 32, 42 °C	published fits (CTMI)	$T_{\max}$: 40–44
$P_{\max}, n$ (Levenspiel)	14% v/v, 1.5	published inhibition kinetics / ASSUMED	n : 0.75–2.25
k_{gly}	1.0 mmol/gDW/h/M	ASSUMED, osmotic physiology	0.5–2.0
NGAM form	0.7 / max(min(f , 1), 0.01)	ASSUMED (Amendment 1)	via n/k_{gly} arms
f_{floor} (viability)	0.01	declared	–
glucose uptake cap	–20 mmol/gDW/h	declared modeling choice	–
oxygen uptake cap	0.5 mmol/gDW/h	declared modeling choice	–
collapse threshold	rel. yield < 0.20	locked definition	–
G2 threshold	fraction $\geq$ 0.25	locked gate	–
solver noise band	10^{-14}	declared	–

Seeds: LHS 20260926; DESIGN/TEST split 424242; screen40 557; design150 558; test150 559; FVA environments 777; bootstrap 20260927; random baseline 30 seeds.

C Solver-timeout register

Sixteen GLPK native hangs occurred campaign-wide (affected designs include the YCL052C knockout and several GPP/PDC/MEP/FPS1/ALD6/TPS scalings). Python-level signals do not reach the solver in this state; each evaluation therefore ran under a hard per-design worker-termination timeout, and every hang is recorded as `solver_timeout` in the stage artifacts. No timed-out evaluation was imputed, retried to a different answer, or silently dropped from a count.

D Judge archive

Two independent-judge rounds were run under the program’s evaluation protocol (20 weaknesses + 20 strengths format) through the owner’s ChatGPT account, archived verbatim with provenance in `docs/JUDGE_ROUNDS.md`; the round-2 verdict text is preserved byte-for-byte in `docs/JUDGE_ROUND2_VERDICT.txt` (SHA-256 1d850952...550cf1d; conversation link and prompt hash recorded alongside). The round-2 verdict (FAIL - ISEF Grand Awards level, on the pre-redirect record) and its reasoning are quoted in the repository and were inputs to the redirect this paper executes. The judge’s three suggested additions - a regulatory layer, anchored validation, and a predictive test - correspond to the outlook items in Section 5.

E Repository map

`src/yeasttwin/` - model loading (SBML pin), stress layers (`climate/parameters.py`, `climate/apply.py`), stage drivers. `results/climate/` - per-stage artifacts with embedded hashes (a1–a8, sweeps, a5/, a6/ run logs). `docs/` - pre-registration and amendments, judge rounds, courier record, standing rules. `data/raw/SECTION8_MANIFEST.json` - dataset accession hashes. `paper/climate.tex` - this document.